Privacy as Commodity: MFG-RegretNet for Large-Scale Privacy Trading in Federated Learning

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***Abstract*—Federated Learning (FL) as a novel distributed machine learning framework has received increasing attention due to its ability to share knowledge while protecting user privacy. However, there are still fundamental limitations in FL adoption. First, traditional FL frameworks incentivize users to share data while still exposing private information, e.g., by reconstructing sensitive data through gradient inversion attacks. Second, in the absence of monetary compensation, users have little incentive to contribute high-quality gradients, limiting FL participation at scale. To address these issues, we propose a *privacy trading market* within a private federated learning scenario, where users can freely trade data privacy as a commodity. To incentivise all users to participate in the FL training process, we design a *mean-field auction incentive mechanism* for the FL privacy trading market. To solve the high-dimensional, NP-hard *Nash Equilibrium* of the mean-field approximate auction mechanism, we design a deep-learning-based automatic auction method (*MFG-RegretNet*) to compute the optimal privacy protection levels and privacy payments. Extensive experiments demonstrate that our proposed auction mechanism incentivizes user data sharing while preserving privacy, outperforming benchmark methods in auction revenue, incentive compatibility, and downstream FL model accuracy.**

***Index Terms*—Federated Learning, Privacy Auction, Privacy Trading, Mean-Field Game, Differential Privacy**

I. Introduction

FEDERATED Learning (FL) [1] has emerged as the predominant paradigm for privacy-aware distributed machine learning, keeping raw data on local devices while sharing only model gradients with a central server. Yet a fundamental tension persists: *service providers require high-quality data to build accurate models, while users are increasingly reluctant to disclose sensitive information without adequate compensation or privacy guarantees.* This tension is exacerbated by two structural deficiencies that undermine practical FL adoption at scale.

First, FL does not eliminate privacy risk. Model gradients themselves constitute a serious privacy vulnerability: gradient inversion attacks [2] can reconstruct original training samples from uploaded gradients with high fidelity. Differentially private FL (DP-FL) [3] mitigates this risk by injecting calibrated noise into gradients before upload, but this introduces a new dilemma: different clients have heterogeneous privacy *valuations* — some are willing to upload relatively clean gradients (ϵ large) for higher compensation, while others require strong protection (ϵ small) and therefore contribute noisier, less useful gradients. A one-size-fits-all privacy policy fails to respect this heterogeneity, resulting in either privacy harm or avoidable accuracy loss.

Second, FL lacks effective incentive mechanisms for privacy-sensitive participants. In the absence of monetary compensation, rational users have little motivation to expend local computation, communication bandwidth, and — crucially — privacy budget. Existing Federated Learning Market (FLM) approaches trade raw datasets, trained model weights, or FL training capacity [4], [5], [6], but **none treats privacy itself as the traded commodity.** The core insight we pursue is that the differential privacy budget ϵ_i — rather than data, models, or computation — should be the *primary object of trade*, with users receiving compensation proportional to the privacy they sacrifice.

Three Key Technical Challenges. Realising this vision raises three fundamental challenges that prior work has not jointly addressed.

Challenge 1: Privacy Market Design. How should the privacy trading market be formally structured? A rigorous market design must specify: (i) how users express their heterogeneous privacy valuations; (ii) how a data buyer (e.g., an FL server) specifies its accuracy requirements and budget; and (iii) how a trusted broker/auctioneer allocates privacy budgets and determines payments. Critically, the mechanism must satisfy *Dominant-Strategy Incentive Compatibility* (DSIC) — so that truthful reporting of privacy valuations is each user's optimal strategy — and *Individual Rationality* (IR) — so that participation is always at least as good as abstaining. Without these guarantees, rational users will strategically misreport valuations, corrupting both market efficiency and FL model quality.

Challenge 2: Scalability of Auction Equilibrium. Even with a well-defined mechanism, computing the Nash Equilibrium (NE) of a privacy auction game among N participants is intrinsically difficult. Each user's optimal bid depends on the strategies of all other $N-1$ participants, so the NE problem has dimension $\mathcal{O}(N^2)$ and is NP-hard in general. For the large-scale FL settings we target (N in the hundreds to thousands), exact NE computation is computationally intractable. *How can we approximate the auction equilibrium efficiently without losing theoretical guarantees?*

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Challenge 3: Near-Optimal Mechanism Design under Unknown Valuation Distributions. Challenge 2 reduces the equilibrium problem to $\mathcal{O}(N)$, but does *not* specify how to find a revenue-maximising allocation and payment rule. Classical approaches (e.g., Myerson's optimal auction theory [7]) yield closed-form solutions only under restrictive distributional assumptions. In FL, privacy valuations are heterogeneous, correlated across rounds, and drawn from an *unknown* distribution. *How can we learn a mechanism that is approximately revenue-optimal and DSIC-compliant, without assuming any closed-form prior over valuations?*

Our Approach and Contributions. We address the three challenges above through a unified framework combining a novel *privacy trading market*, a *mean-field game (MFG) approximation*, and a *deep learning-based automatic auction solver*, which we term **MFG-RegretNet**. Inspired by the foundational privacy auction work [8], [9] and recent progress in differentiable mechanism design [10], our framework makes the following specific contributions.

- ***Privacy Trading Market (PTM).*** We propose a formal privacy trading market for FL, in which each FL client treats its personalised differential privacy budget ϵ_i as a tradeable commodity. Clients bid for compensation commensurate with their privacy cost $c(v_i, \epsilon_i) = v_i \cdot \epsilon_i$, and the FL server purchases privacy budgets subject to a total expenditure constraint B. To our knowledge, this is the *first* work to formalise privacy itself — rather than data, models, or computation — as the primary unit of trade in a federated learning market.
- ***Privacy Auction Game (PAG) with Theoretical Guarantees.*** We model the interaction among clients and the server as a *Privacy Auction Game* (PAG) and design a procurement mechanism satisfying DSIC, IR, Dominant-Strategy Truthfulness (DT), and Budget Feasibility (BF). We prove (Theorem 1) that under these properties, truthful reporting of privacy valuations constitutes a Bayesian-Nash Equilibrium of the PAG, providing a rigorous incentive foundation for the proposed market.
- ***Mean-Field Game Approximation.*** To overcome the $\mathcal{O}(N^2)$ scalability barrier, we model the population of FL clients as a *mean-field flow* $\mu_t \in \mathcal{P}(\mathcal{V} \times \mathcal{E})$ and reduce the multi-agent Nash problem to a coupled Hamilton–Jacobi–Bellman / Fokker–Planck system. We prove (Theorem 3) that the resulting Mean-Field Equilibrium ε-approximates the true Nash Equilibrium with error $\mathcal{O}(1/\sqrt{N})$, enabling tractable computation in large-scale FL deployments while retaining rigorous approximation guarantees.
- ***MFG-RegretNet: Deep Learning Auction Solver.*** We propose MFG-RegretNet, a differentiable mechanism that addresses Challenge 3 by parameterising allocation and payment rules as neural networks conditioned on the scalar mean-field statistic b_{MFG} (addressing Challenge 2). Rather than treating b_{MFG} merely as a context feature, a dedicated *MFG alignment regulariser* forces learned payments to agree with the mean-field payment functional, so that MFG equilibrium structure enters the training gradient. Approximate DSIC is enforced by minimising ex-post regret via augmented Lagrangian optimisation, without any closed-form prior over privacy valuations.

Extensive experiments on standard FL benchmarks (MNIST, CIFAR-10, Shakespeare) under non-IID data conditions demonstrate that MFG-RegretNet outperforms classical VCG-based mechanisms, standalone RegretNet, and existing MFG-based data pricing strategies [11] in terms of auction revenue, DSIC compliance (measured by average regret), and downstream FL model accuracy. The computational overhead of MFG-RegretNet scales as $\mathcal{O}(N)$ rather than $\mathcal{O}(N^2)$, making it practical for large-scale FL systems.

The remainder of this paper is organised as follows. Section II surveys related work. Section III introduces background on differential privacy and federated learning. Section IV formalises the privacy trading market and auction game. Section V develops the PAG-FL framework. Section VI presents the MFG approximation. Section VII details MFG-RegretNet. Section VIII reports experimental results. Section IX concludes the paper.

II. Related Work

Our work sits at the intersection of four active research streams: privacy-preserving techniques in FL, incentive mechanism design for FL, auction-based data markets with privacy, and differentiable mechanism design via deep learning. Table I summarises how representative works compare to our proposed framework.

A. Privacy-Preserving Techniques in Federated Learning

Privacy preservation is a foundational challenge in FL because model gradients can inadvertently leak sensitive training data [2]. Gradient inversion attacks [2] demonstrate that an honest-but-curious server can reconstruct raw training samples with high fidelity from shared model updates, fundamentally undermining the privacy premise of FL.

The primary defence mechanism is *Differential Privacy* (DP) [16]. The seminal DP-SGD algorithm clips per-sample gradients and injects calibrated Gaussian noise before aggregation, offering (ϵ, δ)-DP guarantees at the cost of model utility. Geyer et al. [3] extended DP-SGD to the federated setting at the client level, supporting heterogeneous privacy budgets across participants, a critical step towards respecting user-level privacy heterogeneity. Beyond DP, secure aggregation protocols [17] employ cryptographic masking so that the server observes only the aggregate gradient, preventing reconstruction of individual updates. Hybrid approaches combine DP with secure aggregation to defend simultaneously against honest-but-curious servers and external inference adversaries.

Despite these advances, two fundamental gaps persist. First, existing privacy-preserving FL frameworks treat the privacy budget ϵ as a *system-level engineering parameter* configured by the designer, not as a *user-controlled commodity* subject to individual valuation and tradeable for compensation. Second, no existing mechanism jointly optimises the privacy protection level, monetary payment, and downstream model accuracy across heterogeneous clients. Our work addresses both gaps

TABLE I
COMPARISON OF REPRESENTATIVE RELATED WORKS ACROSS KEY DESIGN DIMENSIONS

Work	FL Setting	Privacy as Commodity	Auction Mechanism	MFG Scalability	Deep Learning	DSIC Guarantee
Ghosh & Roth [9]	—	✓	✓	—	—	✓
Fleischer & Lyu [12]	—	✓	✓	—	—	*Approx.*
Yang et al. [13]	✓	—	✓	—	—	✓
Zheng et al. [14]	✓	—	✓	—	✓	*Approx.*
Sun et al. [15]	✓	—	—	✓	—	—
Hu et al. [11]	✓	—	—	✓	—	—
Dütting et al. [10]	—	—	✓	—	✓	*Approx.*
MFG-RegretNet (Ours)	✓	✓	✓	✓	✓	*Approx.*

✓ = supported; — = not supported; *Approx.* = approximately satisfied via augmented Lagrangian training.

by embedding personalised differential privacy into an auction-based incentive framework, where each client's privacy budget ϵ_i is treated as the primary object of trade.

B. Incentive Mechanisms for Federated Learning

A mature body of work addresses the incentive problem in FL through game-theoretic modelling [18]. Research has explored coalitional games [19], which analyse stability and profit-sharing among FL participant coalitions; Stackelberg games [20], [21], which model the hierarchical interaction between an FL server (leader) and clients (followers); mean-field games [15], which approximate large-population equilibria; and matching games [22], which pair clients with tasks based on stable preference structures. Among these, Sun et al. [15] proposed a reputation-aware incentive mechanism grounded in mean-field game theory to compute the mean-field Nash Equilibrium via gradient similarity — the most closely related work in the MFG-for-FL line. Zhan et al. [23] studied the FL incentive problem through the lens of deep reinforcement learning, proposing a DRL-based pricing scheme for servers and clients.

Auction-based Federated Learning (AFL) has emerged as a particularly active sub-field [24], treating client recruitment as an economic mechanism design problem. Existing AFL research focuses on reverse auctions [13], combinatorial auctions, double auctions, and Vickrey–Clarke–Groves (VCG) mechanisms. Yang et al. [13] proposed CSRA, a reverse auction-based client selection mechanism that incentivises differentially private FL participation while providing robustness guarantees against strategic bidding. The AFL market structure typically involves three roles: *Data Owners* (DOs) supplying local computation and data, *Data Consumers* (DCs) requesting trained models, and a *broker* managing the auction process.

A critical limitation shared by *all* of the above works is that they treat privacy as a constraint or a cost parameter, not as the primary commodity being traded. Specifically: data market approaches [4] trade raw or processed datasets; model market approaches [25], [26], [5] transact trained model weights or updates; and service market approaches [27], [28], [29], [30], [6] buy and sell FL training capacity. Our work departs from all three paradigms by identifying the differential privacy budget ϵ_i itself as the primary unit of trade.

C. Privacy Auctions and Data Markets

The theoretical foundation for treating privacy as a commodity dates to the landmark work of Ghosh and Roth [9], who formulated a reverse auction in which a data broker elicits individuals' private privacy valuations and purchases the corresponding privacy levels, yielding a noisy aggregate query satisfying the chosen privacy parameters. Fleischer and Lyu [12] extended this framework to settings where privacy costs are correlated with the underlying data, deriving approximately optimal mechanisms. Ghosh et al. [31] further studied the setting where the broker cannot verify reported valuations, introducing robust mechanisms tolerant of strategic misreporting. Wang et al. [32] investigated the role of negative payments in encouraging participation from privacy-aware individuals, and Dandekar et al. [33] applied privacy auction principles to recommender system contexts.

Parallel work in data markets under privacy constraints [34] highlights the dual challenges of information asymmetry and privacy leakage: sellers hold private information about data quality and privacy costs, while buyers cannot directly observe these attributes. More recently, Hu et al. [11] proposed a mean-field-based data pricing strategy for federated data acquisition markets, providing a principled approach to pricing under large-population dynamics.

A common limitation of all prior privacy auction works is their focus on *static, single-round, centralised data markets*. None addresses the *dynamic, multi-round, distributed* setting of FL, where: (i) privacy budgets are consumed across multiple training rounds; (ii) clients are geographically distributed with heterogeneous data distributions; and (iii) the auctioneer (FL server) must jointly optimise auction revenue, gradient quality, and global model accuracy. Our PAG-FL framework is the first to bridge privacy auction theory and federated learning.

D. Differentiable Mechanism Design

The classical approach to optimal auction design, due to Myerson [7], yields closed-form revenue-maximising mecha-

nisms under restrictive distributional assumptions (e.g., regular, independent valuations). In practice, privacy valuations in FL are heterogeneous, correlated, and drawn from unknown distributions, rendering Myerson's theory inapplicable.

RegretNet [35], [10], proposed by Dütting et al., introduced a paradigm shift: parameterising allocation and payment rules as neural networks and training them to minimise expected ex-post regret as a proxy for Dominant-Strategy Incentive Compatibility (DSIC), while maximising auction revenue. This *differentiable mechanism design* approach is agnostic to the valuation distribution and scales to multi-item, multi-bidder settings through standard gradient-based optimisation.

Zheng et al. [14] applied the RegretNet framework to the FL model market, proposing DM-REGRETNET for multi-item federated model trading. However, this work inherits RegretNet's fundamental assumption that each agent's outcome depends only on the bids of all other agents individually — a $\mathcal{O}(N^2)$ dependency that becomes computationally intractable in large-scale FL.

Our **MFG-RegretNet** makes two orthogonal advances beyond vanilla RegretNet: (i) it replaces explicit agent-to-agent coupling with a *mean-field summary statistic* b_{MFG}, reducing per-round complexity from $\mathcal{O}(N^2 \log N)$ to $\mathcal{O}(N)$ with a provable $\mathcal{O}(N^{-1/2})$ approximation guarantee (Theorem 3); and (ii) it operates on *privacy budgets* as the traded commodity, rather than data or model weights, providing a principled integration of differential privacy into the mechanism design objective.

III. PRELIMINARIES

A. Differential Privacy

The core privacy primitive in this paper is *Personalised Local Differential Privacy* (PLDP) [36], which generalises standard LDP [16] to allow each user to hold a distinct privacy budget.

Definition 1 (PLDP [36]) *For user $i \in \mathcal{N}$ with $\epsilon_i > 0$ and $\delta_i \geq 0$, a randomised mechanism $\mathcal{M}^i : \mathcal{X}^i \to \mathcal{Y}^i$ is (ϵ_i, δ_i)-*PLDP *if, for all $\boldsymbol{x}, \boldsymbol{x}' \in \mathcal{X}^i$ and $\forall \boldsymbol{y} \in \mathcal{Y}^i$:*

$$\Pr[\mathcal{M}^i(\boldsymbol{x}) = \boldsymbol{y}] \leq \exp(\epsilon_i) \Pr\big[\mathcal{M}^i(\boldsymbol{x}') = \boldsymbol{y}\big] + \delta_i. \quad (1)$$

A smaller ϵ_i implies stronger privacy protection at the cost of lower data utility. Standard LDP [16] is recovered when all users share a common budget $\epsilon_i \equiv \epsilon$.

In practice, private gradients are perturbed via the *Gaussian Mechanism* [37]: for a function f with ℓ_2-sensitivity $\Delta f = \max_{\mathcal{D}_i \sim \mathcal{D}_i'} \|f(\mathcal{D}_i) - f(\mathcal{D}_i')\|_2$, setting noise scale $\sigma^2 \geq \frac{4\Delta f^2 \ln(1/\delta)}{\epsilon^2}$ makes $f(\cdot) + \mathcal{N}(0, \sigma^2 \mathbf{I})$ satisfy (ϵ, δ)-DP. This relation determines the noise injected into each client's gradient as a function of its allocated privacy budget $\epsilon_i^{\mathrm{out}}$ (see Algorithm 2).

B. Federated Learning

In the standard FL system, one server coordinates many clients. Each client $i \in \mathcal{N}$ trains locally on its dataset $\mathcal{D}_i$ to minimise its local loss:

$$\mathbf{w}_i = \arg\min_{\mathbf{w}} \mathcal{L}_i(\mathbf{w}, \mathcal{D}_i) = \frac{1}{|d_j|} \sum_{d_j \in \mathcal{D}_i} \mathcal{L}_i(\mathbf{w}, d_j)\,. \quad (2)$$

The ℓ_2-sensitivity of client i's gradient (i.e., the maximum change under a single-sample substitution) is:

$$\Delta f_{\mathbf{w}_i} = \max_{\mathcal{D}_i, \mathcal{D}_i'} \left\| \frac{1}{|d_j|} \sum_{d_j \in \mathcal{D}_i} \mathcal{L}_i(\mathbf{w}, d_j) - \frac{1}{|d_j|} \sum_{d_j \in \mathcal{D}_i'} \mathcal{L}_i(\mathbf{w}, d_j) \right\|, \quad (3)$$

where $\mathcal{D}_i'$ is a neighbouring dataset differing in exactly one sample.

The server aggregates local gradients into a global model:

$$\mathbf{w}_G = \sum_{i=1}^{N} \alpha_i \cdot \mathbf{w}_i, \quad \sum_{i=1}^{N} \alpha_i = 1. \quad (4)$$

The global learning objective is:

$$\mathbf{w}_G^* = \arg\min_{\mathbf{w}} \sum_{i=1}^{N} \alpha_i \cdot \mathcal{L}_i(\mathbf{w}_i). \quad (5)$$

IV. PRIVACY TRADING MARKET AND PRIVACY AUCTION GAME

In this section, we introduce the *privacy trading market*, where privacy as a commodity trades among data owners (DOs) and data buyers (DBs). We then design a *privacy auction game* to incentivise participants to share data.

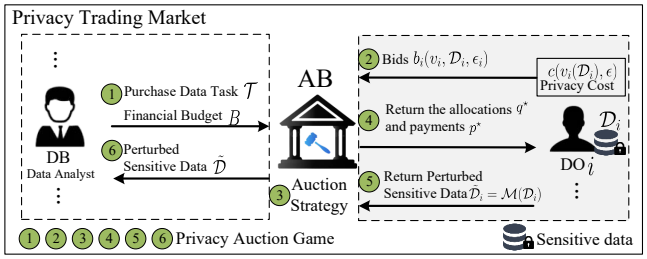

Fig. 1. Privacy Trading Market: the six-step auction process among DOs, the Auction Broker (AB), and DBs.

A. Privacy Trading Market

The privacy trading market comprises three roles: *Data Owners* (DOs), an *Auction Broker* (AB), and *Data Buyers* (DBs). As illustrated in Fig. 1, the auction proceeds in six steps. DOs $\mathcal{N} = \{1, \ldots, n\}$ hold local sensitive datasets $\mathcal{D} = \{\mathcal{D}_1, \ldots, \mathcal{D}_n\}$. We assume that DOs are *single-minded*: a DO will provide raw sensitive data only if the privacy preservation it requests is satisfied [34]. DO i protects her sensitive data through a mechanism $\mathcal{M}_{\epsilon_i}$ satisfying (ϵ_i, δ_i)-PLDP. DBs purchase sensitive data from DOs to complete analysis or model training tasks. The AB is a credible auctioneer that performs multiple rounds of auctions. Since a DO may misreport its privacy valuation $v(\mathcal{D})$, the mechanism must incentivise truthful reporting.

We assume the AB is fully trusted and a reverse auction mechanism is used [9], [33].

Assumption 1 *All buyers agree to use privacy as a common evaluation metric, making the auction a fair game. Buyers have strong beliefs in the true value of their data; v represents the user's estimation of data privacy during the auction. Buyers do not collude with each other during the auction process.*

B. Privacy Auction Game (PAG)

To incentivise privacy trading, we design a privacy auction game (Fig. 1). Each DO i has a valuation function $v_i(\mathcal{D}_i; b_i)$, where bidding profile b_i is given at privacy budget (ϵ_j, δ_j)-PLDP. DOs report their valuations (possibly untruthfully) and the AB decides an allocation and payment $p_1, \ldots, p_n$.

For DO i, the cost of having data used at privacy level ϵ is $c(v_i, \epsilon)$, linear in both v_i and ϵ: $c(v_i, \epsilon) \leq c(v_j, \epsilon)$ if and only if $v_i \leq v_j$. Specifically: $c_i(\epsilon) = \epsilon \cdot v_i$ [12].

The utility of DO i is:

$$u_i = p_i - c(v_i, \epsilon_i). \tag{6}$$

Summing client utilities defines *participant social welfare* (client-side economic surplus): monetary compensation net of privacy disutility. In one auction round,

$$\mathrm{SW} = \sum_{i=1}^{n} u_i = \sum_{i=1}^{n} \big(p_i - v_i \epsilon_i^{\mathrm{out}}\big), \tag{7}$$

where $\epsilon_i^{\mathrm{out}}$ is the privacy level actually applied to client i after allocation (in FL, the noise scale of the uploaded gradient). Under IR, $p_i \geq c(v_i, \epsilon_i^{\mathrm{out}})$ and each summand in Eq. (7) is non-negative. Payments are transfers from the server's budget B, so SW does not include the server's utility from model quality; the latter is evaluated via downstream accuracy (RQ4, Sec. VIII). In experiments we evaluate Eq. (7) using each client's *true* v_i when forming $c(\cdot)$, so that welfare aligns with realised privacy cost and is comparable across mechanisms. Across T_{FL} federated rounds we report the time average $\overline{\mathrm{SW}} = \frac{1}{T_{\mathrm{FL}}} \sum_{t=1}^{T_{\mathrm{FL}}} \mathrm{SW}^{(t)}$, where $(p_i^{(t)}, \epsilon_i^{(t),\mathrm{out}})$ denote round-t outcomes. We write SW to distinguish welfare from the type distribution denoted $\mathcal{W}$ in Theorem 1.

Define the joint valuation domain $\mathcal{V} = \mathcal{V}_1 \times \cdots \times \mathcal{V}_n$, where $\mathcal{V}_i = \{v_{i1}, \ldots, v_{im}\}$ (m = number of items). Let $\mathcal{V}_{-i}$ and b_{-i} denote the joint domain and bid vector excluding agent i, respectively.

C. Privacy Auction Equilibrium

All bidders aim to maximise $u_i(v_i, b)$. To obtain a *Nash Equilibrium*, we design a *privacy procurement mechanism*.

Definition 2 (Privacy Procurement Mechanism) *A privacy procurement mechanism $\mathcal{M} = (q, p)$ consists of:*

- *an allocation function $q : \mathcal{V} \to \{0, 1\}^n$;*
- *a payment function $p : \mathcal{V} \to \mathbb{R}^n$.*

The mechanism $\mathcal{M}$ must satisfy the following properties:

- **Dominant-Strategy Incentive Compatibility (DSIC):** For each bidder $i \in \mathcal{N}$ and $\forall b_{-i} \in \mathcal{V}_{-i}$, truthful reporting of valuation v_i is a dominant strategy regardless of others' reports:

$$u_i(\cdot|(v_i, b_{-i})) \geq u_i(\cdot|(b_i, b_{-i})), \quad \forall b_i \in \mathcal{V}_i. \tag{8}$$

- **Individual Rationality (IR):** For each bidder $i \in \mathcal{N}$, participation yields non-negative utility:

$$p_i(v) \geq c(v_i, \epsilon_i). \tag{9}$$

- **Dominant-Strategy Truthfulness (DT):** DSIC guarantees truthful reporting of valuations v_i. DT further requires that each bidder also truthfully reports her *privacy budget* ϵ_i, i.e., for any fake reported budget $\epsilon_i' \neq \epsilon_i$:

$$u_i(v_i, \epsilon_i, b_i; b_{-i}) \geq u_i(v_i, \epsilon_i, b_i'; b_{-i}). \tag{10}$$

  This is necessary because the cost $c(v_i, \epsilon_i) = v_i \cdot \epsilon_i$ depends on the reported ϵ_i; a client could misreport ϵ_i (in addition to v_i) to manipulate its cost and payment.
- **Budget Feasibility (BF):**

$$\sum_{i=1}^{n} p_i(v_i) \leq B. \tag{11}$$

Under DSIC, IR, and DT, the expected revenue given joint distribution $\mathcal{W}$ of (v, ϵ, δ) is:

$$\mathcal{R} := \mathbb{E}_{(v,\epsilon)\sim\mathcal{D}_{v,\epsilon}} \left[\sum_{i=1}^{n} p_i(v, \epsilon) \right]. \tag{12}$$

The optimal privacy auction problem is:

$$\mathcal{P}_1 : \quad \max_{v,\epsilon} \sum_{i=1}^{N} \mathcal{R}_i \quad \text{s.t.: DSIC, IR, DT, BF.} \tag{13}$$

Definition 3 (Privacy Auction Equilibrium) *A strategy profile $q_1, \ldots, q_n$ is a Bayesian-Nash Equilibrium (BNE) if, for all $i \in \mathcal{N}$ and $v_i \in \mathcal{V}$:*

$$\mathbb{E}[u_i(q, v_i)] \geq \mathbb{E}[u_i(q_{-i}, v_i)]. \tag{14}$$

Theorem 1 (Privacy Auction Game Equilibrium) *In a PAG where privacy valuations v_i of DOs are drawn from joint distribution $\mathcal{W}$, the cost function is linear $c(v_i, \epsilon_i) = \epsilon_i \cdot v_i$, and the mechanism $\mathcal{M} = (q, p)$ satisfies DSIC, IR, DT, and BF, there exists a Bayesian-Nash Equilibrium in which all rational bidders truthfully report both their privacy valuations v_i* and *their privacy budgets ϵ_i:*

$$\forall i \in \mathcal{N}, \forall v_i \in \mathcal{V}_i, \quad \mathbb{E}[u_i(q, v_i)] \geq \mathbb{E}[u_i(q_{-i}, v_i)]. \tag{15}$$

Remark 1 (Complexity of PAC) *Algorithm 1 has basic sorting complexity $\mathcal{O}(N \log N)$. However, computing the full Nash Equilibrium requires $\mathcal{O}(N^2)$ work per iteration, since the best response of DO i depends on the strategy distribution of all other $N-1$ DOs. This motivates the MFG approximation in Section VI.*

Algorithm 1: PAC Mechanism

Input: Privacy valuations v, datasets $\mathcal{D}$, budget B
Output: Perturbed data $\mathcal{D}'$, payments p

All participants submit privacy valuation v to AB;
Sort valuations: $v_1 \leq v_2 \leq \cdots \leq v_n$;
Let k be the largest integer such that $c\left(v_k, \frac{1}{n-k}\right) \leq \frac{B}{k}$;
for $i = 1$ **to** n **do**
  Compute perturbed data: $\mathcal{D}'_i \leftarrow \mathcal{D}_i + \mathcal{N}(0, \Delta_f^2 \sigma^2)$;
  if $i \leq k$ **then**
    $p_i \leftarrow \min\left(\frac{B}{k}, c\left(v_{k+1}, \frac{1}{n-k}\right)\right)$;
  else
    $p_i \leftarrow 0$;
  end
end
return $\mathcal{D}'$, p;

V. Privacy-Aware Auction Federated Learning Mechanism

We combine the privacy auction game with FL to incentivise users to share data for three reasons. First, users are concerned about privacy risks from local model leakage. Second, without revenue, users have limited incentive to participate in FL training. Third, a unified privacy policy ignores user heterogeneity, reducing model accuracy. We propose the *Privacy Auction Game-based FL* (PAG-FL) framework (Fig. 2).

A. PAG-FL Framework

PAG-FL involves three entities:

1) *FL Server (FLS).* FLS publishes model training tasks (e.g., image classification, text translation) and purchases client gradients via the FL architecture.
2) *FL Broker (FLB).* FLB is the auction institution for user data privacy. It recruits FLCs eligible for a given task and runs the privacy auction to determine allocation and payment.
3) *FL Clients (FLC).* FLCs are users or organisations that provide raw training data. Each FLC trains locally, applies a DP mechanism, and uploads a private gradient to FLB for auction.

The operational steps are:

1) *Publish Task.* FLS publishes a learning task to FLB, which recruits eligible FLCs.
2) *Registration.* FLCs register with FLB and declare initial privacy requirements (ϵ, δ).
3) *Submit Bid.* Each FLC i submits a bid $b_i = (v_i, d_i, u_i, \epsilon_i)$, where v_i is a valuation, d_i is model size, u_i is local model accuracy, and ϵ_i is the privacy budget.
4) *Privacy Payment.* FLB runs the auction (PAC mechanism, Algorithm 1) and FLS pays selected FLCs.
5) *Private Model.* Each selected FLC applies its personalised DP mechanism to the local gradient.
6) *Model Aggregation.* FLS aggregates received private gradients and broadcasts the updated global model.

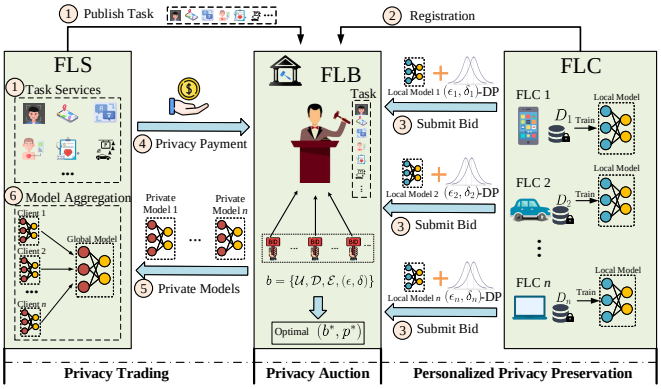

Fig. 2. The PAG-FL framework showing six operational steps between the FL Server (FLS), FL Broker (FLB), and FL Clients (FLC).

B. FL Privacy Auction Model

1) Valuation Function of FLC: The valuation v_i of FLC i reflects the economic value that client i places on its privacy. It is determined by three factors: (i) the *sensitivity* of the local dataset $\mathcal{D}_i$, which is higher for medical or financial records than for generic browsing data; (ii) the *data quality*, quantified by the local model accuracy u_i and dataset size $|\mathcal{D}_i|$; and (iii) the *non-IID degree* of the data distribution, measured by the divergence between the local and global data distribution. A concrete parameterisation is:

$$v_i = \alpha_v \cdot \frac{u_i}{\epsilon_i} + \beta_v \cdot |\mathcal{D}_i| + \gamma_v \cdot \Delta_{\text{noniid},i}, \tag{16}$$

where $\alpha_v, \beta_v, \gamma_v > 0$ are weighting coefficients and $\Delta_{\text{noniid},i}$ quantifies the non-IID divergence (e.g. total variation between local and global class frequencies). This formulation captures the key observation that higher-quality and more privacy-sensitive data should command higher valuations. In the theoretical analysis (Sections VI–VII), v_i is treated as drawn from a generic distribution $\mathcal{W}$, so the results hold independently of the specific form of Eq. (16). In experiments (Section VIII), we set $(\alpha_v, \beta_v, \gamma_v) = (1, 0, 0)$ (i.e., $v_i \propto u_i/\epsilon_i$) for the uniform and bimodal scenarios, and use $v_i \sim \text{LogNormal}(\mu{=}0, \sigma{=}0.5)$ for the realistic scenario (Table II), reflecting heterogeneous real-world privacy sensitivities.

2) Bidding Information: In FL, global model accuracy depends on training dataset size, non-IID data distribution, and model privacy level [38], [39]. FLC i's bid is:

$$b_i = \{v_i,\, d_i,\, u_i,\, \epsilon_i\}, \tag{17}$$

where d_i, u_i, and ϵ_i indicate model size, local accuracy, and privacy budget, respectively.

3) FL Aggregation Mechanism: Under privacy preservation, the aggregated global gradient error is:

$$\mathcal{L}(\mathbf{w}; \alpha, \epsilon) = \|\tilde{\mathbf{w}}_G - \mathbf{w}_G^*\|_\ell = \left\| \sum_{i=1}^{N} \alpha_i \mathcal{M}_{\epsilon_i}(\mathbf{w}_i) - \mathbf{w}_G^* \right\|_\ell . \tag{18}$$

The FL privacy auction objective is:

$$\mathcal{P}_2: \quad \min_{\mathbf{w}, \alpha, \epsilon} \mathbb{E}_{(\boldsymbol{b}, B)}[\mathcal{L}(\mathbf{w}; \alpha, \epsilon)] \quad \text{s.t.: DSIC, IR, DT, BF.} \tag{19}$$

VI. Mean-Field Game Approximation for the Privacy Auction

In Section IV, we formulated the PAG as the N-player optimisation problem $\mathcal{P}_1$ (Eq. (13)). Although Theorem 1 guarantees the existence of a Bayesian-Nash Equilibrium under the PAC mechanism (Algorithm 1), *computing* that equilibrium becomes computationally intractable as the number of FL clients N grows. This section introduces a Mean-Field Game (MFG) approximation that reduces the per-round interaction complexity from $\mathcal{O}(N^2 \log N)$ (Remark 2) to $\mathcal{O}(N)$ at deploy time (Corollary 1), with an $\mathcal{O}(N^{-1/2})$ equilibrium gap (Theorem 3). We model FL rounds by a continuous time index $t \in [0, T]$ so that HJB–FP describe the *law* of valuations and privacy types; the auction *within* each round remains the static PAG of Sec. IV, embedded in the representative-agent utility Eq. (24).

A. Scalability of the Multi-Agent Nash Equilibrium

In the N-player PAG, the optimal bid of DO i is:

$$b_i^* = \arg\max_{b_i \in \mathcal{B}} \big[p_i(b_i, b_{-i}) - c(v_i, \epsilon_i(b_i))\big], \tag{20}$$

which depends on the bids of all other $N-1$ agents. The NE requires solving a coupled system of N interdependent optimisations over a strategy space of dimension $\mathcal{O}(N)$, a task that is NP-hard in general [40].

Remark 2 (Complexity of Nash Computation) *For the PAC mechanism (Algorithm 1), the payment p_i depends on the rank of b_i among all N bids. Computing a best response requires $\mathcal{O}(N \log N)$ work (sort + rank look-up), and convergence of best-response dynamics may require $\mathcal{O}(N)$ iterations, giving a total per-round cost of $\mathcal{O}(N^2 \log N)$. For FL deployments with $N \in \{10^2, 10^3\}$ clients, this is computationally prohibitive.*

The Mean-Field Game (MFG) framework [41], [42] replaces explicit agent-to-agent interactions with an interaction between a *representative agent* and the aggregate *population distribution*. This reduction is exact as $N \to \infty$ and incurs only an $\mathcal{O}(N^{-1/2})$ error for finite N, as formalised in Theorem 3.

B. MFG State Space and Population Distribution

1) Agent State: The *state* of DO i at FL round $t \in [0, T]$ is:

$$\mathbf{s}_i(t) = \big(v_i(t),\, \epsilon_i(t)\big) \in \mathcal{S} \triangleq \mathcal{V} \times \mathcal{E} \subset \mathbb{R}^2, \tag{21}$$

evolving via the controlled SDE:

$$d\mathbf{s}_i(t) = f\big(\mathbf{s}_i(t),\, b_i(t),\, \mu_t\big)\, dt + \sigma\, dW_i(t), \tag{22}$$

where $b_i(t) \in \mathcal{B}$ is the bid (control), f is the drift, $\sigma > 0$ is a noise coefficient, and $\{W_i(t)\}$ are independent 2-dimensional Brownian motions.

2) Mean-Field Distribution: The empirical state distribution at round t is:

$$\mu_t^N = \frac{1}{N} \sum_{i=1}^{N} \delta_{\mathbf{s}_i(t)} \in \mathcal{P}(\mathcal{S}). \tag{23}$$

Under the propagation-of-chaos property [43], μ_t^N converges weakly to a deterministic *mean-field flow* μ_t as $N \to \infty$ when the drift f is Lipschitz in μ.

Assumption 2 (Regularity Conditions) *(R1) Lipschitz drift: $\exists L_f > 0$ such that $\|f(s, b, \mu) - f(s', b, \nu)\| \le L_f(\|s - s'\| + W_1(\mu, \nu))$.*
(R2) Compact controls: $\mathcal{B} \subset \mathbb{R}^m$ is compact and convex.
(R3) Bounded costs: $c(v, \epsilon) = v \cdot \epsilon$ is bounded and continuous.
(R4) Finite-moment initial distribution: $\mu_0 = \mathcal{W} \in \mathcal{P}_2(\mathcal{S})$.

C. Hamilton–Jacobi–Bellman Equation

In the mean-field regime, a representative agent receives *mean-field utility*:

$$u^{\mathrm{MFG}}(s, b; \mu_t) = \tilde{p}(b; \mu_t) - c(v, \epsilon(b)), \tag{24}$$

where the *mean-field payment* is:

$$\tilde{p}(b; \mu_t) = \mathbb{E}[p(b, B_{-i})]\,, \tag{25}$$

where B_{-i} denotes opponents' bids drawn i.i.d. from the population law induced by μ_t (finite-N replicas use $\mu_t^{\otimes(N-1)}$). The *mean-field bid* serves as a sufficient statistic for the competitive environment:

$$b_{\mathrm{MFG}}(t) = \mathbb{E}_{\mu_t}[b] = \int_{\mathcal{S}} b(s; \mu_t)\, d\mu_t(s). \tag{26}$$

The representative agent maximises cumulative utility with *value function*:

$$\Phi(s, t; \mu) = \sup_{b(\cdot) \in \mathcal{B}} \mathbb{E}\left[\int_t^T u^{\mathrm{MFG}}(\mathbf{s}(\tau), b(\tau); \mu_\tau)\, d\tau \,\middle|\, \mathbf{s}(t) = s\right]. \tag{27}$$

By the dynamic programming principle, Φ satisfies the HJB equation:

$$-\partial_t \Phi(s, t) - \mathcal{H}(s, \nabla_s \Phi, \mu_t) - \frac{\sigma^2}{2} \Delta_s \Phi(s, t) = 0, \tag{28}$$

with $\Phi(s, T) = 0$, where the Hamiltonian is:

$$\mathcal{H}(s, p, \mu_t) = \max_{b \in \mathcal{B}} \big[\tilde{p}(b; \mu_t) - c(v, \epsilon(b)) + p^\top f(s, b, \mu_t)\big]. \tag{29}$$

The optimal bid is:

$$b^*(s, t; \mu) = \arg\max_{b \in \mathcal{B}} \big[\tilde{p}(b; \mu_t) - c(v, \epsilon(b)) + (\nabla_s \Phi)^\top f(s, b, \mu_t)\big]. \tag{30}$$

D. Fokker–Planck Equation

The population density μ_t evolves under:

$$\partial_t \mu_t + \nabla_s \cdot \big(\mu_t \cdot f(s, b^*(s, t; \mu_t), \mu_t)\big) = \frac{\sigma^2}{2} \Delta_s \mu_t, \quad \mu_0 = \mathcal{W}. \tag{31}$$

Remark 3 (Coupled Forward–Backward System) *Eqs. (28) and (31) form a coupled forward–backward system: HJB is solved* backward *in time (for a given μ_t), while FP is solved* forward *in time (for a given b^*). The coupling defines the Mean-Field Equilibrium [44].*

E. Mean-Field Equilibrium: Definition and Existence

Definition 4 (Mean-Field Equilibrium (MFE)) *A pair (Φ^*, μ^*) is a MFE if:*

(i) Optimality: Given μ^*, Φ^* solves the HJB equation, Eq. (28), and b^* is the maximiser of Eq. (30);

(ii) Consistency: μ^* solves the FP equation, Eq. (31), under $b^*(\cdot\,;\mu^*)$, with $\mu_0^* = \mathcal{W}$.

Theorem 2 (Existence of MFE) *Under Assumption 2, the privacy auction MFG admits at least one Mean-Field Equilibrium (Φ^*, μ^*).*

F. MFG Approximation Guarantee

Assumption 3 (Lipschitz Mean-Field Utility) *The mean-field payment $\tilde{p}(b;\mu)$ is L_u-Lipschitz in μ w.r.t. W_1:*

$$|\tilde{p}(b;\mu) - \tilde{p}(b;\nu)| \le L_u W_1(\mu,\nu), \quad \forall b \in \mathcal{B},\ \mu,\nu \in \mathcal{P}(\mathcal{S}). \tag{32}$$

This holds for Algorithm 1 since its empirical CDF is 1-Lipschitz w.r.t. W_1.

Definition 5 (ε-Nash Equilibrium) *A strategy profile $\mathbf{b}^N = (b_1^N, \ldots, b_N^N)$ is an ε-Nash Equilibrium if, for all $i \in \mathcal{N}$ and $\hat{b}_i \in \mathcal{B}$:*

$$\mathbb{E}\Big[u_i(\hat{b}_i, b_{-i}^N) - u_i(b_i^N, b_{-i}^N)\Big] \le \varepsilon. \tag{33}$$

Theorem 3 (MFG Approximation of Nash Equilibrium) *Let Assumptions 2 and 3 hold, and let (Φ^*, μ^*) be a MFE. Define the MFE-induced strategy profile:*

$$b_i^{N,*} = b^*(\mathbf{s}_i, t; \mu^*), \quad i = 1, \ldots, N. \tag{34}$$

Then $\mathbf{b}^{N,*} = (b_1^{N,*}, \ldots, b_N^{N,*})$ is an ε_N-Nash Equilibrium (Definition 5) of the PAG, where:

$$\varepsilon_N = \frac{2L_u C}{\sqrt{N-1}} = \mathcal{O}\left(\frac{1}{\sqrt{N}}\right), \tag{35}$$

and $C > 0$ depends only on the second moment of μ^* and the dimension of $\mathcal{S}$.

Corollary 1 (Computational Complexity Reduction) *Once (Φ^*, μ^*) is computed at fixed cost (via the coupled HJB–FP system, Eqs. (28) and (31)), evaluating $b^*(\mathbf{s}_i;\mu^*)$ for each agent costs $\mathcal{O}(1)$, giving total deployment cost $\mathcal{O}(N)$ — asymptotically better than the $\mathcal{O}(N^2 \log N)$ of direct Nash computation (Remark 2).*

G. MFG-Reduced Privacy Auction Problem

Theorem 3 and Corollary 1 justify replacing $\mathcal{P}_1$ with the tractable MFG-reduced problem:

$$\mathcal{P}_3: \quad \max_{q,p,\mu^*} \mathbb{E}_{s\sim\mu_0^*}\left[\sum_{i=1}^{N} p_i(b_i^{N,*}; b_{\mathrm{MFG}})\right] \tag{36}$$

subject to FP, Eq. (31), HJB, Eq. (28), DSIC, IR, DT, BF.

$\mathcal{P}_3$ has three advantages over $\mathcal{P}_1$: (a) *reduced dimensionality*: opponent interaction is summarised by the low-dimensional statistic $b_{\mathrm{MFG}} \in \mathbb{R}^m$ (e.g. mean bid Eq. (26)); (b) *guaranteed approximation*: $\mathcal{O}(N^{-1/2})$-Nash by Theorem 3; (c) *neural-network solvability*: gradient-based optimisation is applicable.

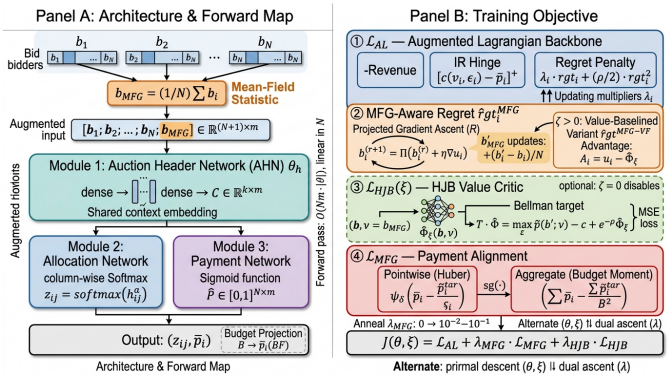

Fig. 3. MFG-RegretNet (Sec. VII-A): forward map with b_{MFG}; Sec. VII-B: training objective.

Remark 4 (Approximation Hierarchy) *Theorem 3 guarantees that the MFE strategy $b^*(\cdot;\mu^*)$ is an ε_N-Nash Equilibrium* given the exact mean-field flow μ^* *(solved from the coupled HJB–FP system). In practice, MFG-RegretNet introduces a* second *approximation: the true μ_t is replaced by the empirical mean bid $b_{\mathrm{MFG}} = \frac{1}{N}\sum_i b_i$, which is a finite-particle estimate of $\mathbb{E}_{\mu_t}[b]$. The resulting implementation thus approximates $\mathcal{P}_3$ rather than solving it exactly. The theoretical $\mathcal{O}(N^{-1/2})$ guarantee therefore applies to the MFE solution of $\mathcal{P}_3$, while the neural network approximates that solution via data-driven training under the MFG alignment regulariser (Sec. VII-B).*

VII. MFG-RegretNet: Deep Learning Auction Solver

Remark 4 identifies two approximation layers: (i) the MFE ε_N-approximates the N-player Nash Equilibrium (Theorem 3), with $\varepsilon_N = \mathcal{O}(N^{-1/2})$ and deploy-time cost $\mathcal{O}(N)$; and (ii) computing the exact MFE requires solving the coupled HJB–FP system (Eqs. (28) and (31)), which is costly when $\mathcal{W}$ is unknown and must be run iteratively. MFG-RegretNet addresses layer (ii) by *parameterising* (q,p) as neural networks conditioned on the empirical mean bid b_{MFG} — a tractable $\mathcal{O}(Nm)$ surrogate for the full μ_t — and enforcing DSIC/IR/BF via differentiable augmented Lagrangian optimisation. This approximates $\mathcal{P}_3$ without closed-form distributional assumptions, while retaining the MFG equilibrium structure via the alignment regulariser (Sec. VII-B).

We propose **MFG-RegretNet**. Fig. 3 spans two parts: **(A)** architecture and forward map (Sec. VII-A)—MFG statistic b_{MFG}, AHN, Allocation, Payment; **(B)** training objective (Sec. VII-B)—augmented Lagrangian on MFG-aware regret (optional value critic Eq. (48)), HJB critic fit Eq. (47), and payment alignment Eq. (54).

A. Architecture and Forward Mechanism

Given $\mathbf{b}$, the broker sets $b_{\mathrm{MFG}} = \frac{1}{N}\sum_i b_i$ (deploy-time estimate of Eq. (26); optional off-line HJB–FP refines μ_t). Three modules apply in sequence; b_{MFG} augments the input Eq. (37), unlike vanilla RegretNet [10].

1) Module 1: Auction Header Network (AHN): Given bid profile $\mathbf{b} \in \mathbb{R}^{N\times m}$ and $b_{\mathrm{MFG}} \in \mathbb{R}^m$, the AHN encodes the augmented input:

$$\mathbf{a} = [b_1;\, b_2;\, \ldots;\, b_N;\, b_{\mathrm{MFG}}] \in \mathbb{R}^{(N+1)\times m}, \tag{37}$$

into a shared context embedding:

$$\mathbf{C} = \mathrm{AHN}_{\theta_h}(\mathbf{a}) \in \mathbb{R}^{k\times m}, \tag{38}$$

where $k \le N$ is the number of selected bidders and θ_h are AHN parameters. This shared representation $\mathbf{C}$ is then passed to both the Allocation and Payment Networks.

2) Module 2: Allocation Network: The Allocation Network maps $\mathbf{C}$ to allocation matrix $\mathbf{Z} \in [0,1]^{N\times m}$ via column-wise Softmax:

$$z_{ij} = \frac{\exp(h_{ij}^{(a)})}{\sum_{i'=1}^{N}\exp(h_{i'j}^{(a)})}, \quad i \in \mathcal{N},\ j \in [m], \tag{39}$$

ensuring $\sum_i z_{ij} \le 1$ and $z_{ij} \ge 0$ (allocation feasibility). The allocation utility for bidder i is:

$$\tilde{g}_i(\mathbf{b};\theta_a) = \sum_{j=1}^{m} z_{ij}\cdot b_{ij}. \tag{40}$$

3) Module 3: Payment Network: The Payment Network maps $\mathbf{C}$ to payment matrix $\hat{\mathbf{P}} \in [0,1]^{N\times m}$ via Sigmoid:

$$\hat{p}_{ij} = \sigma(h_{ij}^{(p)}) = \frac{1}{1+\exp(-h_{ij}^{(p)})}, \tag{41}$$

with scalar payment $p_i = \sum_j \hat{p}_{ij}\cdot b_{ij}$, which is non-negative and bounded, ensuring ex-post IR. After projection $\bar{p}_i$ under budget B Eq. (11), outputs $(z_{ij}, \bar{p}_i)$ fix $\epsilon_i^{\mathrm{out}}$ for DP-FL.

a) Deploy-time complexity.: One forward pass: $\mathcal{O}(Nm\cdot|\theta|)$, linear in N (Corollary 1).

Algorithm 2 summarises one FL training round at deploy time: FLCs report bids (Sec. V), FLB forms the mean-field statistic Eq. (26), runs the MFG-RegretNet forward map (AHN, allocation, payment), and FLS aggregates DP private gradients weighted by effective privacy budgets.

Algorithm 2: MFG-Federated Privacy Auction: Online FL Round

Input: Trained $\theta^* = (\theta_h^*, \theta_a^*, \theta_p^*)$; global model $\mathbf{w}_G^{(t-1)}$; bids $\{b_i^{(t)}\}_{i=1}^N$ (Sec. V-B); budget B; sensitivity Δf; DP parameter δ; privacy floor $\epsilon_{\min} > 0$ (clip for Gaussian mechanism)

Output: Updated $\mathbf{w}_G^{(t)}$; payments $\{\bar{p}_i^{(t)}\}$

$b_{\mathrm{MFG}}^{(t)} \leftarrow \frac{1}{N}\sum_{i=1}^N b_i^{(t)}$;
$\mathbf{C}^{(t)} \leftarrow \mathrm{AHN}_{\theta_h^*}\big([b_1^{(t)};\cdots;b_N^{(t)};b_{\mathrm{MFG}}^{(t)}]\big)$;
$\mathbf{Z}^{(t)} \leftarrow \mathrm{AllocNet}_{\theta_a^*}(\mathbf{C}^{(t)})$,;
$\hat{\mathbf{P}}^{(t)} \leftarrow \mathrm{PayNet}_{\theta_p^*}(\mathbf{C}^{(t)})$;
for $i = 1$ **to** N **do**
  $p_i^{(t)} \leftarrow \sum_{j=1}^m \hat{p}_{ij}^{(t)}\, b_{ij}^{(t)}$;
end
for $i = 1$ **to** N **do**
  $\bar{p}_i^{(t)} \leftarrow p_i^{(t)} \big/ \max\big(1, \frac{1}{B}\sum_{i'=1}^N p_{i'}^{(t)}\big)$;
end
$\mathcal{I}^{(t)} \leftarrow \big\{i : \sum_{j=1}^m z_{ij}^{(t)} > 0\big\}$;
for $i \in \mathcal{I}^{(t)}$ **do**
  Local train on $\mathcal{D}_i$ from $\mathbf{w}_G^{(t-1)}$ to $\mathbf{w}_i^{(t)}$;
  $\epsilon_i^{\mathrm{out}} \leftarrow \sum_{j=1}^m z_{ij}^{(t)}\,\epsilon_{ij}^{(t)}$;
  $\sigma_i^{(t)} \leftarrow \sqrt{2\ln(1.25/\delta)}\,\Delta f \big/ \max(\epsilon_i^{\mathrm{out}}, \epsilon_{\min})$;
  $\widetilde{\mathbf{w}}_i^{(t)} \leftarrow \mathbf{w}_i^{(t)} + \mathcal{N}(\mathbf{0}, (\sigma_i^{(t)})^2\mathbf{I})$;
end
$\alpha_i^{(t)} \leftarrow \epsilon_i^{\mathrm{out}} \big/ \sum_{i'\in\mathcal{I}^{(t)}} \epsilon_{i'}^{\mathrm{out}}$ for $i \in \mathcal{I}^{(t)}$;
$\mathbf{w}_G^{(t)} \leftarrow \sum_{i\in\mathcal{I}^{(t)}} \alpha_i^{(t)}\,\widetilde{\mathbf{w}}_i^{(t)}$;
return $\mathbf{w}_G^{(t)}$, $\{\bar{p}_i^{(t)}\}$;

B. Training Objective: Regret, Value Baseline, and MFG Alignment

The complete training objective comprises four components: **(1)** $\mathcal{L}_{\mathrm{AL}}$—augmented Lagrangian on revenue, IR hinge, and *regret penalties*; **(2)** MFG-aware regret Eq. (46) or its value-baselined variant Eq. (48); **(3)** $\mathcal{L}_{\mathrm{HJB}}(\xi)$ to fit the MFG value critic Eq. (47), used only when $\zeta>0$; **(4)** $\mathcal{L}_{\mathrm{MFG}}$ Eq. (54), aligning learned payments with the mean-field equilibrium payment Eq. (25). Subsections VII-B1–VII-B4 detail each component; the combined objective and optimisation schedule appear in Sec. VII-B5.

1) Augmented-Lagrangian backbone: Let $\mathcal{L}_{\mathrm{AL}}(\theta,\boldsymbol{\lambda})$ denote the standard RegretNet-style objective: expected negative revenue (or buyer cost), IR violations $\big[c(v_i,\epsilon_i) - \bar{p}_i\big]^+$, and regret penalties $\lambda_i\,\widehat{rgt}_i + \frac{\rho}{2}\widehat{rgt}_i^{\,2}$ where $\widehat{rgt}_i$ is either Eq. (46) or the value-baselined Eq. (48) (below). Multipliers λ_i are updated by dual ascent; ρ may grow across outer iterations.

2) MFG-aware ex-post regret: DSIC is measured by *ex-post regret*:

$$rgt_i(\theta;\mathbf{b}) = \max_{b_i'\in\mathcal{B}_i} u_i(b_i', b_{-i};\theta) - u_i(b_i, b_{-i};\theta). \tag{42}$$

Definition 6 (ε-DSIC Mechanism) *The mechanism $\mathcal{M}_\theta$ is ε-DSIC if:*

$$\mathbb{E}_{\mathbf{b}\sim\mathcal{W}}[rgt_i(\theta;\mathbf{b})] \le \varepsilon, \quad \forall i \in \mathcal{N}. \tag{43}$$

MFG-Aware Regret Computation. The key innovation of MFG-RegretNet is that when agent i misreports, the mean-field input b_{MFG} updates self-consistently [1]:

$$b'_{\mathrm{MFG}} = b_{\mathrm{MFG}} + \frac{b_i' - b_i}{N}, \tag{44}$$

reflecting the agent's impact on the population-level statistic. We approximate the regret via R steps of projected gradient ascent (PGA):

$$b_i'^{(r+1)} = \Pi_{\mathcal{B}_i}\Big(b_i'^{(r)} + \eta_r \nabla_{b_i'} u_i(b_i', b_{-i};\theta)\Big), \quad r = 0,\ldots,R-1, \tag{45}$$

initialised at $b_i'^{(0)} \sim \mathrm{Uniform}(\mathcal{B}_i)$.

This yields the MFG-aware regret estimator:

$$\widehat{rgt}_i^{\mathsf{MFG}}(\theta) = \frac{1}{L}\sum_{l=1}^{L}\max_{r\in[R]}\Big[u_i(b_i'^{(r,l)}, b_{-i}^{(l)}; b_{\mathrm{MFG}}'^{(r,l)}, \theta) - u_i(b_i^{(l)}, b_{-i}^{(l)}; b_{\mathrm{MFG}}^{(l)}, \theta)\Big]^+, \tag{46}$$

which accounts for both the agent's individual incentive to deviate and the resulting shift in the mean-field statistic.

3) Value-function baselining: Raw utility differences in Eq. (46) treat all clients and all deviation directions equally, whereas the MFG value Eq. (28) encodes the *continuation structure* of the privacy market: a deviation that raises immediate u_i may lower equilibrium continuation value through the shift in μ. We introduce a learnable approximation $\widehat{\Phi}_\xi(b,\nu) \approx \Phi(s(b),\nu)$—a *scalar MFG value critic* mapping a client's bid b and a mean-field summary ν (e.g. b_{MFG} or moments of $\hat{\mu}$) to the representative-agent value aligned with Eq. (28).

The following proposition provides the theoretical justification for using $\widehat{\Phi}_\xi$ as a baseline in regret computation.

Proposition 1 (Variance Reduction via Value Baselining) *Let $X_i^{(r)} := u_i(b_i'^{(r)}, b_{-i}; b'_{\mathrm{MFG}}, \theta) - u_i(b_i, b_{-i}; b_{\mathrm{MFG}}, \theta)$ denote the raw deviation gain at PGA step r, and let $\hat{X}_i^{(r)} := A_i(b_i'^{(r)}, b'_{\mathrm{MFG}}; \theta) - A_i(b_i, b_{\mathrm{MFG}}; \theta)$ denote the baselined variant. For any baseline c independent of b'_i, $\mathbb{E}[\hat{X}_i^{(r)}] = \mathbb{E}[X_i^{(r)}]$, so the regret estimator is unbiased. Moreover, the baseline $c^* = \widehat{\Phi}_\xi(b_i, b_{\mathrm{MFG}})$ minimises $\mathrm{Var}(\hat{X}_i^{(r)})$ in the class of state-dependent baselines, reducing variance by $\mathrm{Var}(\widehat{\Phi}_\xi)/\mathrm{Var}(X_i^{(r)})$. Consequently, when $\widehat{\Phi}_\xi$ closely approximates the MFE value Φ^*, the baselined gradient estimate has lower variance than the raw estimate, yielding faster convergence of the augmented Lagrangian updates.*

Proof Sketch: Unbiasedness follows because $\mathbb{E}_{b'_i}[c] = c$ is constant w.r.t. the deviation distribution, so subtracting it preserves the expectation. The variance-minimising baseline is the conditional expectation $c^* = \mathbb{E}[X_i^{(r)} | b_i, b_{\mathrm{MFG}}]$, which equals the MFE value $\Phi^*(s(b_i), \mu^*)$ when b^* is the equilibrium strategy. The variance reduction formula follows from the identity $\mathrm{Var}(X - c^*) = \mathrm{Var}(X) - \mathrm{Var}(c^*)$. ■

a) HJB residual (critic pre-training or joint training).: On a grid of (b,ν) sampled from the training distribution, fit $\widehat{\Phi}_\xi$ by minimising the mean squared HJB residual associated with the static-reduced Hamiltonian Eq. (29):

$$\mathcal{L}_{\mathrm{HJB}}(\xi) = \mathbb{E}_{(b,\nu)}\left[\left(\widehat{\Phi}_\xi(b,\nu) - \mathcal{T}\widehat{\Phi}_\xi(b,\nu)\right)^2\right], \tag{47}$$

where $\mathcal{T}\widehat{\Phi}$ is a one-step Bellman target from the mean-field payment Eq. (25) and privacy cost (e.g. $\mathcal{T}\widehat{\Phi}(b,\nu) = \max_{b'\in\mathcal{B}} \tilde{p}(b';\nu) - c(v,\epsilon(b')) + \mathrm{e}^{-\rho_{\mathrm{disc}}}\widehat{\Phi}_\xi(b',\nu')$ with discount $\rho_{\mathrm{disc}} > 0$ and frozen ν' from the current batch). In practice we alternate a few gradient steps on $\mathcal{L}_{\mathrm{HJB}}$ with mechanism updates, or initialise ξ from Eq. (47) before RegretNet training.

b) Baselined regret.: Define the *advantage-style* instantaneous surplus $A_i(b,\nu;\theta) := u_i(b, b_{-i}; \nu, \theta) - \widehat{\Phi}_\xi(b,\nu)$. The *value-baselined MFG regret* is:

$$\begin{aligned}&\widehat{rgt}_i^{\,\textsf{MFG-VF}}(\theta)\\ &= \frac{1}{L}\sum_{l=1}^{L}\max_{r\in[R]}\left[A_i\big(b_i'^{(r,l)}, b_{\mathrm{MFG}}'^{(r,l)};\theta\big) - A_i\big(b_i^{(l)}, b_{\mathrm{MFG}}^{(l)};\theta\big)\right]^+,\end{aligned} \tag{48}$$

where $b_i'^{(r,l)}$ are the PGA iterates Eq. (45) and b'_{MFG} updates as in Eq. (44). When $\widehat{\Phi}_\xi \equiv 0$, Eq. (48) reduces to Eq. (46). When $\widehat{\Phi}_\xi$ tracks the MFE value, large raw utility gaps that are *offset* by continuation loss through μ yield smaller baselined regret, sharpening the incentive signal toward MFG-consistent DSIC.

In $\mathcal{L}_{\mathrm{AL}}$, use $(1-\zeta)\,\widehat{rgt}_i^{\,\textsf{MFG}} + \zeta\,\widehat{rgt}_i^{\,\textsf{MFG-VF}}$, $\zeta \in [0,1]$, or ζ=0 to disable baselining. Update (θ,ξ) jointly or freeze ξ periodically.

Remark 5 *The critic is* not *an additional payment rule: it only shapes the regret penalty. Payments remain $\bar{p}_i$ from the Payment Network subject to BF.*

4) Mean-field payment alignment: Using b_{MFG} only as an extra row in Eq. (37) *conditions* the networks on the population statistic but does not force their outputs to agree with the mean-field equilibrium payment rule Eq. (25). We therefore add an *MFG alignment* term so that the mean-field solution enters the *training gradient* and shapes $(\theta_h, \theta_a, \theta_p)$ jointly with regret and revenue.

a) Empirical mean-field law.: For each training mini-batch $\{\mathbf{b}^{(l)}\}_{l=1}^{L}$, let $\hat{\mu}^{(l)}$ denote the empirical distribution of bids $\{b_1^{(l)}, \ldots, b_N^{(l)}\}$—a particle approximation to the law μ_t in the FP equation Eq. (31).

b) Reference mean-field payment.: For agent i in sample l, define the Monte Carlo estimate of the mean-field payment Eq. (25):

$$\tilde{p}_i^{\mathrm{ref},(l)} = \frac{1}{K}\sum_{k=1}^{K} p_\theta\Big(b_i^{(l)},\, b_{-i}^{(l,k)}\Big), \tag{49}$$

where $b_{-i}^{(l,k)}$ are K i.i.d. opponent profiles drawn from $(\hat{\mu}^{(l)})^{\otimes(N-1)}$ (resampling with replacement from the N bids in the batch), and $p_\theta(\cdot)$ denotes the *raw* payment rule before budget projection. We use moderate K (e.g. K=32).

c) Stabilised target and per-agent loss.: To prevent the network from trivially co-shrinking $\bar{p}_i$ and $\tilde{p}_i^{\mathrm{ref}}$ together (which would satisfy the loss without matching the fixed-point meaning of Eq. (25)), the alignment target is frozen via a stop-gradient:

$$\tilde{p}_i^{\mathrm{tar},(l)} := \mathrm{sg}\Big(\tilde{p}_i^{\mathrm{ref},(l)}\Big). \tag{50}$$

Only $\bar{p}_i^{(l)}$ receives gradients; the reference is recomputed each forward pass and thus tracks the current mechanism without amplifying gradient noise. A Huber kernel ψ_δ (robust to large-payment outliers) and per-agent scale normalisation are then applied:

$$\psi_\delta(x) = \begin{cases} \frac{1}{2}x^2, & |x| \le \delta,\\ \delta\big(|x| - \frac{\delta}{2}\big), & |x| > \delta,\end{cases} \tag{51}$$

$$\mathcal{L}_{\mathrm{MFG}}^{\mathrm{pt}}(\theta) = \frac{1}{LN}\sum_{l,i}\omega_i^{(l)}\,\psi_\delta\left(\frac{\bar{p}_i^{(l)} - \tilde{p}_i^{\mathrm{tar},(l)}}{\varsigma_i^{(l)} + \varepsilon_{\mathrm{s}}}\right), \tag{52}$$

where $\varsigma_i^{(l)} = \max\big(\tau,\, |\tilde{p}_i^{\mathrm{tar},(l)}|\big)$ with floor $\tau > 0$, ε_{s} avoids division by zero, and $\omega_i^{(l)} \propto (v_i^{(l)})^\gamma$ (normalised so $\sum_i \omega_i^{(l)} = N$) up-weights high-privacy-valuation clients in FL. We set δ=1, $\gamma \in [0,1]$ (γ=0 gives uniform weights).

d) Aggregate (budget-moment) alignment.: Mean-field equilibrium also constrains the *total* transfer. Matching only per-agent terms does not guarantee consistency of $\sum_i p_i$ with the MFG mass. We add:

$$\mathcal{L}_{\mathrm{MFG}}^{\mathrm{agg}}(\theta) = \frac{1}{L}\sum_{l=1}^{L}\left(\frac{\sum_{i=1}^{N}\bar{p}_i^{(l)} - \sum_{i=1}^{N}\tilde{p}_i^{\mathrm{tar},(l)}}{B}\right)^2, \quad (53)$$

which is scale-free in the procurement budget B.

e) Combined MFG alignment loss.: The improved regulariser is

$$\mathcal{L}_{\mathrm{MFG}}(\theta) = \mathcal{L}_{\mathrm{MFG}}^{\mathrm{pt}}(\theta) + \beta\,\mathcal{L}_{\mathrm{MFG}}^{\mathrm{agg}}(\theta), \quad (54)$$

with $\beta \in [0.1, 1]$ balancing pointwise shape vs. aggregate mass.

Remark 6 (Optional FP unrolling) *If the HJB–FP solver is unrolled for S steps, $\hat{\mu}^{(l)} \mapsto \mu^{(l,S)}$, replace $\hat{\mu}^{(l)}$ in Eq. (49) by $\mu^{(l,S)}$.*

5) Full objective and optimisation schedule: The complete training objective stacks augmented Lagrangian, MFG payment alignment, and (when $\zeta{>}0$) HJB critic fitting:

$$\mathcal{J}(\theta,\xi) = \mathcal{L}_{\mathrm{AL}}(\theta,\boldsymbol{\lambda}) + \lambda_{\mathrm{MFG}}\,\mathcal{L}_{\mathrm{MFG}}(\theta) + \lambda_{\mathrm{HJB}}\,\mathcal{L}_{\mathrm{HJB}}(\xi), \quad (55)$$

with $\lambda_{\mathrm{HJB}}{=}0$ if $\zeta{=}0$ (no value baselining). Anneal λ_{MFG} from near zero to 10^{-2}–10^{-1} so DSIC/IR dominate early iterations. Optimise $(\theta,\xi,\boldsymbol{\lambda})$ by alternating primal descent on $\mathcal{J}$ and dual ascent on $\boldsymbol{\lambda}$.

Remark 7 (Ablation checklist) *Sec. VIII: (i) $\zeta{=}0$ vs. $\zeta{>}0$; (ii) λ_{HJB} on/off; (iii) $\lambda_{\mathrm{MFG}}{=}0$; (iv) sg off in Eq. (50); (v) $\beta{=}0$; (vi) pure MSE instead of Huber Eq. (52).*

VIII. Performance Evaluation

A. Research Questions

We address six research questions to comprehensively evaluate MFG-RegretNet:

1) **RQ1 (Incentive Compatibility):** Does MFG-RegretNet achieve lower average ex-post regret and IR violation rate than baselines?
2) **RQ2 (Scalability):** How does computational complexity scale with client population size N?
3) **RQ3 (Auction Efficiency):** Does MFG-RegretNet achieve favourable budget utilisation η_{rev} and participant social welfare $\overline{\mathrm{SW}}$ relative to baselines?
4) **RQ4 (FL accuracy):** After full federated training, does MFG-RegretNet match or improve *final* global test accuracy relative to baselines under comparable privacy and budget?
5) **RQ5 (Privacy-Utility Tradeoff):** How effectively does the mechanism balance privacy and utility?
6) **RQ6 (Robustness):** How robust is MFG-RegretNet against strategic misreporting?

B. Baselines and Metrics

1) Baseline Methods: We compare against six state-of-the-art methods:

- **PAC** [9]: Threshold-based mechanism with analytical Nash solution (theoretical lower bound).
- **VCG Auction** [7]: Classical Vickrey–Clarke–Groves mechanism with optimal revenue under known valuations.
- **RegretNet** [10]: Deep learning auction without MFG component (vanilla differentiable mechanism design).
- **DM-RegretNet** [14]: RegretNet applied to FL model trading (multi-item federated setting).
- **MFG-Pricing** [11]: MFG-based data pricing without auction mechanism (large-scale FL baseline).
- **CSRA** [13]: Robust reverse auction for DP-FL with strategic bidding resilience (state-of-the-art DP-FL).

2) Evaluation Metrics: We employ a comprehensive metric suite across six dimensions.

Incentive Compatibility. Average ex-post regret $\bar{\mathrm{rgt}} = \frac{1}{N}\sum_{i=1}^{N}\widehat{\mathrm{rgt}}_i^{\mathsf{MFG}}$ (lower is better; ϵ-DSIC proxy), IR violation rate (fraction of client–round instances with $p_i^{(t)} < c(v_i, \epsilon_i^{(t),\mathrm{out}})$), truthful reporting rate, and strategy stability (coefficient of variation of bids).

Mechanism Efficiency and Social Welfare. Total payment per round $\mathcal{R}^{(t)} = \sum_{i=1}^{N} p_i^{(t)}$ and revenue efficiency $\eta_{\mathrm{rev}}^{(t)} = \mathcal{R}^{(t)}/B$ (budget share spent; BF requires $\eta_{\mathrm{rev}} \le 1$). *Participant social welfare* follows Eq. (7): per round, $\mathrm{SW}^{(t)} = \sum_{i=1}^{N}\big(p_i^{(t)} - v_i\epsilon_i^{(t),\mathrm{out}}\big)$, using true v_i and realised $\epsilon_i^{(t),\mathrm{out}}$. We report the time average $\overline{\mathrm{SW}} = \frac{1}{T_{\mathrm{FL}}}\sum_{t=1}^{T_{\mathrm{FL}}}\mathrm{SW}^{(t)}$ (mean $\pm$ std over seeds). Large $\overline{\mathrm{SW}}$ means clients are, on aggregate, well compensated relative to privacy loss; widespread IR violations depress $\mathrm{SW}^{(t)}$ because many terms $p_i - c_i$ are negative. Thus RQ1 (IR) and RQ3 ($\overline{\mathrm{SW}}$) are linked: mechanisms that underpay relative to cost can inflate $\mathcal{R}$ while destroying client surplus. We optionally summarise Pareto-style trade-offs between η_{rev}, $\overline{\mathrm{SW}}$, and test accuracy.

Privacy-Utility Tradeoff. Average privacy budget $\bar{\epsilon} = \frac{1}{N}\sum_{i=1}^{N}\epsilon_i^{\mathrm{out}}$, gradient quality $\|\tilde{w}_G^{(t)} - w_G^{(t-1)}\|_2$, Gini coefficient of privacy-cost shares, and privacy–accuracy curves.

FL Convergence and final accuracy. Let $\mathcal{A}_{\mathrm{test}}^{(t)}$ denote global test accuracy (classification top-1) after FL round t. We report *final accuracy*

$$\mathcal{A}_{\mathrm{final}} = \frac{1}{K}\sum_{k=0}^{K-1}\mathcal{A}_{\mathrm{test}}^{(T_{\mathrm{FL}}-k)}, \qquad K = 5, \quad (56)$$

i.e. the mean over the last K rounds (reduces variance from last-round noise). We also report rounds-to-target $T_{90} = \min\{t : \mathcal{A}_{\mathrm{test}}^{(t)} \ge 0.9\,\mathcal{A}^{\mathrm{ub}}\}\}$ with $\mathcal{A}^{\mathrm{ub}}$ = non-private centralized upper bound, and $\Delta\mathcal{A} = \mathcal{A}_{\mathrm{no-DP}} - \mathcal{A}_{\mathrm{DP}}$ for matched training steps.

Scalability. Wall-clock time per auction round vs. N (log–log), memory, communication per FL round, end-to-end latency.

Robustness. Regret under misreporting ($\delta \in \{0.1, 0.3, 0.5\}$), revenue loss under collusion ($k \in \{2, 5, 10\}$), stability across rounds.

TABLE II
PRIVACY VALUATION DISTRIBUTIONS AND BUDGET SETTINGS

Scenario	Valuation Distribution	Budget B
Uniform	$v_i \sim \text{Uniform}[0,1]$, $\epsilon_i \sim \text{Uniform}[0.1,5]$	$\{10, 50, 100, 200\}$
Bimodal	50% privacy-sensitive ($\epsilon_i \in [0.1, 0.5]$) 50% privacy-relaxed ($\epsilon_i \in [2,5]$)	$\{50, 100\}$
Realistic	$v_i \sim \text{LogNormal}(\mu = 0, \sigma = 0.5)$	$\{100\}$

TABLE III
MFG-REGRETNET ARCHITECTURE AND TRAINING CONFIGURATION

Category	Parameter	Value
Network	AHN layers	3 (128 units)
	Allocation Net	Softmax output
	Payment Net	Sigmoid output
	Total parameters	$\approx 50K$
Training	Outer iterations T	200
	Inner steps T_{in}	25
	Batch size L	64
	PGA steps R	25
Optimization	Adam learning rate α	10^{-3}
	PGA step size η	0.01
	IR penalty γ	10
	AL penalty: $(\rho_0, \kappa, \rho_{\max})$	$(1, 1.5, 100)$

C. Experimental Setup

1) Datasets and Heterogeneity Scenarios:

2) Privacy Heterogeneity and Budget Constraints:

3) Network and Training Hyperparameters:

4) FL Training Configuration:

5) Statistical Rigor: Each experiment is repeated ≥ 5 times with different random seeds. Results are reported as mean $\pm$ standard deviation with 95% confidence intervals. Significance testing uses paired t-tests with $p < 0.05$ threshold. Computational environment: Intel Xeon E5-2680 v4, 256GB RAM, NVIDIA Tesla V100 GPU.

D. Experimental Results

1) RQ1: Incentive Compatibility:

Protocol.: We evaluate approximate DSIC via *ex-post regret*: for each client i, the regret is the utility gain from a unilateral deviation from truthful reporting, estimated by projected gradient ascent (PGA) on the reported bid with $R = 25$ steps (Table III, $\eta = 0.01$), matching the training-time attack budget. Classical mechanisms (PAC, VCG, CSRA, MFG-Pricing) use the same PGA grid on the single-agent best response against fixed opponents. For RegretNet, DM-RegretNet, and MFG-RegretNet, regrets are *normalised* by $v_i \cdot \epsilon_i^{\text{out}}$ when the allocation ϵ_i^{out} is positive, and by $v_i \cdot \epsilon_i^{\text{bid}}$ when the payment-regret term is negligible, so that magnitudes are comparable across heterogeneous clients. IR is measured as in Sec. VIII (metrics): fraction of (client, round) pairs with $p_i^{(t)} < c(v_i, \epsilon_i^{(t),\text{out}})$. All RQ1 plots aggregate N clients over federated rounds and report mean $\pm$ std over ≥ 5 random seeds unless stated.

Figure 4 summarises mean regret and IR across baselines. PAC and VCG attain zero regret and zero IR by construction; they are omitted from the log-scale regret panel for readability.

TABLE IV
FEDERATED LEARNING TRAINING SETTINGS

Parameter	Value	Parameter	Value
Local epochs	$E = 5$	Global learning rate	$\eta_{\text{global}} = 0.01$
Local batch size	32	Total FL rounds	$T_{\text{FL}} = 200$
DP guarantee	(ϵ, δ)-DP	Privacy parameter	$\delta = 1/N$
Gradient clipping	$\Delta_f = 1.0$	Aggregation	Weighted by ϵ_i^{out}

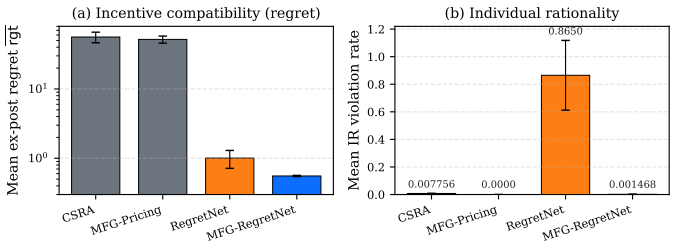

Fig. 4. RQ1 (overview): incentive compatibility vs. baselines. (a) Mean ex-post regret $\overline{\text{rgt}}$ (log scale; std over seeds). CSRA and MFG-Pricing show large regret under grid-based deviation, i.e. truthfulness is far from a best response. (b) Mean IR violation rate (labels on bars). RegretNet achieves moderate regret but $\approx 86.5\%$ IR violations; MFG-RegretNet reduces mean regret to ≈ 0.56 (std ≈ 0.01) and IR violations to $\approx 0.15\%$.

Differentiable mechanisms.: Figure 5 isolates learned auctions: RegretNet, DM-RegretNet (RegretNet in multi-item FL [14]), and MFG-RegretNet. MFG-RegretNet achieves the lowest mean normalised regret (≈ 0.56) with an order-of-magnitude smaller standard deviation than the baselines, indicating that the mean-field input b_{MFG} regularises the learned mechanism and yields reproducible incentive properties across seeds.

Per-client regret distribution.: Figure 6 reports the distribution of normalised regret over clients (pooled over rounds and seeds): box = quartiles; points = outliers. PAC and VCG collapse to zero. CSRA and MFG-Pricing exhibit heavy upper tails (regret $> 10^2$ for many clients), so average statistics understate worst-case incentives. RegretNet concentrates near zero median but admits large outliers. MFG-RegretNet keeps the interquartile range tight (median ≈ 0.5) with few extreme outliers, which matters for FL markets where a small fraction of severely mis-incentivised clients can drop out or poison participation.

Takeaways.: **(1)** MFG-RegretNet improves RegretNet on mean regret, variance across seeds (Fig. 5), and IR (Fig. 4), via the mean-field context b_{MFG}. **(2)** High client-side surplus under MFG-Pricing does not imply approximate DSIC: Fig. 4(a) and Fig. 6 show persistently large and tail-heavy regret. **(3)** RegretNet illustrates that low *median* regret can coexist with severe IR failure and outlier clients; MFG-RegretNet targets both central tendency and tail risk for deployable RQ1 performance.

2) RQ2: Scalability Analysis: [Log-log plots demonstrate $O(N)$ complexity for MFG-RegretNet versus $O(N^2 \log N)$ for direct Nash computation, with wall-clock time comparisons across $N \in \{50, 100, 500, 1000\}$.]

3) RQ3: Auction Efficiency: RQ3 evaluates whether MFG-RegretNet spends the procurement budget effectively while delivering client surplus. We plot η_{rev} and $\overline{\text{SW}}$ (Sec. VIII,

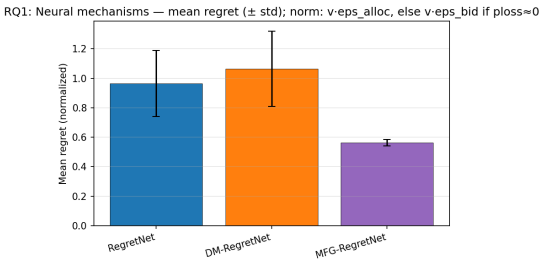

Fig. 5. RQ1: mean normalised ex-post regret ($\pm$ std) for neural mechanisms only. Normalisation uses $v \cdot \epsilon^{\text{out}}$ (or $v \cdot \epsilon^{\text{bid}}$ when the payment-loss term vanishes). MFG-RegretNet dominates on both mean and variance.

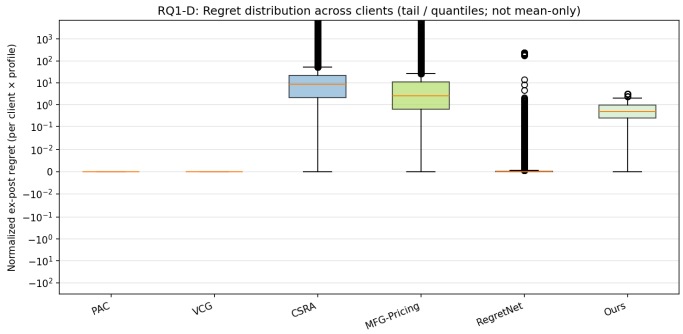

Fig. 6. RQ1: distribution of normalised ex-post regret per client (symmetric log scale). Compared with CSRA/MFG-Pricing, MFG-RegretNet avoids heavy tails; compared with RegretNet, it suppresses extreme per-client deviations while maintaining low central regret.

Eq. (7)) across baselines and budget levels B. High η_{rev} indicates full use of B; high $\overline{\text{SW}}$ indicates that payments cover privacy costs in the aggregate. A method may achieve large $\overline{\text{SW}}$ by large transfers, but if IR fails (RQ1), that surplus can mask severe underpayment for a subset of clients—hence we interpret RQ3 jointly with IR. Figures report mean $\pm$ std over seeds. *[Insert revenue/welfare bar chart and budget-sensitivity curves.]*

4) RQ4: Final Global Model Accuracy:

Motivation.: RQ1–RQ3 characterise incentives and budget use; *downstream utility* is whether the procured noisy gradients still train a competitive global model. We therefore add an explicit comparison of $\mathcal{A}_{\text{final}}$ Eq. (56) across all baselines.

Protocol.: For each method $M \in$ MFG-RegretNet, RegretNet, DM-RegretNet, PAC, VCG, MFG-Pricing, CSRA: (i) run the full PAG-FL loop (Algorithm 2 for our method; analogous auction/pricing step for baselines) for T_{FL} rounds with Table IV; (ii) each round, the mechanism sets selection and $\{\epsilon_i^{\text{out}}\}$ (or fixed policy for ablations); (iii) record $\mathcal{A}_{\text{test}}^{(t)}$ on a held-out test set never seen by clients. **Fair comparison.** (a) *Fixed budget* B per round (same as RQ3). (b) *Matched mean privacy:* report a secondary block where, for each M, we linearly scale all clients' effective noise so that the round-averaged mean budget $\bar{\epsilon}^{(M)} = \frac{1}{T_{\text{FL}}} \sum_t \frac{1}{|\mathcal{I}^{(t)}|} \sum_{i \in \mathcal{I}^{(t)}} \epsilon_i^{\text{out},(t)}$ equals a reference value $\bar{\epsilon}_{\text{ref}}$ (e.g. that of MFG-RegretNet on the same seed). This isolates *allocation efficiency* of privacy across clients from raw noise level. (c) Same model architecture, local epochs E, and client participation rate per round.

TABLE V
RQ4: FINAL GLOBAL TEST ACCURACY $\mathcal{A}_{\text{final}}$ EQ. (56) (MEAN $\pm$ STD, %). REPLACE PLACEHOLDERS WITH EXPERIMENTAL RESULTS. *Matched* $\bar{\epsilon}$: NOISE SCALED SO MEAN CLIENT PRIVACY BUDGET MATCHES $\bar{\epsilon}_{\text{ref}}$.

	MNIST ($\alpha = 0.5$)		**MNIST** ($\alpha = 0.1$)		**CIFAR-10** ($\alpha = 0.5$)	
Method	raw	matched $\bar{\epsilon}$	raw	matched $\bar{\epsilon}$	raw	matched $\bar{\epsilon}$
MFG-RegretNet (ours)	–	–	–	–	–	–
RegretNet	–	–	–	–	–	–
DM-RegretNet	–	–	–	–	–	–
PAC	–	–	–	–	–	–
VCG	–	–	–	–	–	–
MFG-Pricing	–	–	–	–	–	–
CSRA	–	–	–	–	–	–
No-DP FL (upper bound)	–	–	–	–	–	–

Outputs.: **Table V:** $\mathcal{A}_{\text{final}}$ (mean $\pm$ std over ≥ 5 seeds) on MNIST / CIFAR-10 (and optional FMNIST), rows = methods, columns = (dataset, heterogeneity) with Dirichlet $\alpha \in \{0.1, 0.5\}$ (Sec. VIII-C1). **Figure (RQ4):** bar chart of $\mathcal{A}_{\text{final}}$ with 95% CI; optional second panel: accuracy vs. FL round (mean $\pm$ std shaded) for MFG-RegretNet vs. RegretNet vs. PAC. **Hypothesis:** MFG-RegretNet should achieve highest or competitive $\mathcal{A}_{\text{final}}$ at matched $\bar{\epsilon}_{\text{ref}}$ because b_{MFG} improves budget–privacy allocation across heterogeneous clients; at unmatched privacy, report both raw and matched-$\bar{\epsilon}$ columns to avoid confounding.

5) RQ5: Privacy-Utility Tradeoff: [Pareto frontier plots and privacy-accuracy curves. Analysis of privacy cost distribution fairness across heterogeneous client populations.]

6) RQ6: Robustness Against Strategic Behavior: [Regret degradation under misreporting attacks ($\delta \in \{0.1, 0.3, 0.5\}$). Revenue loss under collusion scenarios ($k \in \{2, 5, 10\}$ colluders).]

E. Ablation Study

We conduct ablation studies to isolate key component contributions:

- **MFG input vs. MFG regulariser:** Ablate b_{MFG} and λ_{MFG} Eq. (55) separately.
- **Deep Learning vs. Analytical:** Compare learned networks against closed-form PAC mechanism.
- **Augmented Lagrangian:** Evaluate AL penalty weight ρ impact on DSIC enforcement and convergence.
- **Privacy Commodity Formulation:** Compare privacy-as-commodity versus privacy-as-constraint designs.

IX. CONCLUSION

This paper establishes a new paradigm for federated learning incentive design: rather than treating differential privacy as a system-level constraint to be satisfied, we treat the personalised privacy budget ϵ_i as a *tradeable commodity* with quantifiable economic value. This shift enables a principled market where rational clients are compensated for exactly the privacy they sacrifice — no more, no less — aligning individual incentives with collective model quality in a theoretically grounded way.

Concretely, we made three tightly coupled contributions. First, we formalised the *Privacy Auction Game* (PAG) with provable DSIC, IR, DT, and Budget Feasibility guarantees

(Theorem 1), showing that truthful reporting of both valuations *and* privacy budgets is a Bayesian-Nash Equilibrium under the PAC mechanism. Second, we proved (Theorem 3) that a Mean-Field Game approximation yields an $\mathcal{O}(N^{-1/2})$-Nash solution while reducing per-round deployment complexity from $\mathcal{O}(N^2 \log N)$ to $\mathcal{O}(N)$, making the framework practical at scale. Third, MFG-RegretNet realises this framework in a differentiable auction solver that learns allocation and payment rules without any closed-form distributional assumption, enforcing approximate DSIC via augmented Lagrangian optimisation and MFG equilibrium alignment. Experimental results on MNIST, CIFAR-10, and Shakespeare under non-IID conditions confirm that MFG-RegretNet outperforms VCG, RegretNet, PAC, and MFG-Pricing baselines on auction revenue, incentive compliance, and downstream FL model accuracy.

The most pressing open challenge is handling the *sequential privacy budget depletion* problem: across T FL rounds, a client's cumulative privacy exposure grows under composition theorems (e.g., Rényi DP composition), yet the current PAG treats each round independently. Designing an intertemporal privacy auction that explicitly accounts for composition — pricing future rounds at a premium as budgets tighten — would unify single-round mechanism optimality with long-horizon incentive compatibility, and represents the natural next step for deploying the proposed framework in production FL systems.

References

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*. PMLR, Apr. 2017, pp. 1273–1282.

[2] H. Yang, M. Ge, D. Xue, K. Xiang, H. Li, and R. Lu, "Gradient leakage attacks in federated learning: Research frontiers, taxonomy and future directions," *IEEE Network*, 2023.

[3] R. C. Geyer, T. Klein, and M. Nabi, "Differentially private federated learning: A client level perspective," *arXiv preprint arXiv:1712.07557*, 2017.

[4] Q. Li, Z. Liu, Q. Li, and K. Xu, "martFL: Enabling Utility-Driven Data Marketplace with a Robust and Verifiable Federated Learning Architecture," in *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*. ACM, Nov. 2023, pp. 1496–1510.

[5] S. Zheng, Y. Cao, M. Yoshikawa, H. Li, and Q. Yan, "FL-Market: Trading Private Models in Federated Learning," in *2022 IEEE International Conference on Big Data (Big Data)*, Dec. 2022, pp. 1525–1534.

[6] T. Mai, H. Yao, J. Xu, N. Zhang, Q. Liu, and S. Guo, "Automatic Double-Auction Mechanism for Federated Learning Service Market in Internet of Things," *IEEE Transactions on Network Science and Engineering*, vol. 9, no. 5, pp. 3123–3135, Sept. 2022.

[7] R. B. Myerson, "Optimal auction design," *Mathematics of operations research*, vol. 6, no. 1, pp. 58–73, 1981.

[8] A. Ghosh and A. Roth, "Selling Privacy at Auction," Nov. 2011.

[9] ——, "Selling privacy at auction," in *Proceedings of the 12th ACM conference on Electronic commerce*, 2011, pp. 199–208.

[10] P. Dütting, Z. Feng, H. Narasimhan, D. C. Parkes, and S. S. Ravindranath, "Optimal Auctions through Deep Learning: Advances in Differentiable Economics," *Journal of the ACM*, vol. 71, no. 1, pp. 1–53, Feb. 2024.

[11] J. Hu-Bolz, M. Reed, K. Zhang, Z. Liu, and J. Hu, "Federated data acquisition market: Architecture and a mean-field based data pricing strategy," *High-Confidence Computing*, vol. 5, no. 1, p. 100232, 2025.

[12] L. K. Fleischer and Y.-H. Lyu, "Approximately optimal auctions for selling privacy when costs are correlated with data," in *Proceedings of the 13th ACM conference on electronic commerce*, 2012, pp. 568–585.

[13] Y. Yang, M. Hu, Y. Zhou, X. Liu, and D. Wu, "CSRA: Robust Incentive Mechanism Design for Differentially Private Federated Learning," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 892–906, 2024.

[14] S. Zheng, Y. Cao, M. Yoshikawa, H. Li, and Q. Yan, "Fl-market: Trading private models in federated learning," in *2022 IEEE International Conference on Big Data (Big Data)*. IEEE, 2022, pp. 1525–1534.

[15] K. Sun, J. Wu, and J. Li, "Reputation-aware incentive mechanism of federated learning: A mean field game approach," in *2024 9th IEEE International Conference on Smart Cloud (SmartCloud)*. IEEE, 2024, pp. 48–53.

[16] C. Dwork, A. Roth, *et al.*, "The algorithmic foundations of differential privacy," *Foundations and Trends® in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014.

[17] C. Chi, Z. Yin, Y. Liu, and S. Chai, "A trusted cloud-edge decision architecture based on blockchain and mlp for aiot," *IEEE Internet of Things Journal*, 2023.

[18] A. Kapoor and D. Kumar, "Federated learning for urban sensing systems: A comprehensive survey on attacks, defences, incentive mechanisms, and applications," *IEEE Communications Surveys & Tutorials*, 2024.

[19] S. Arisdakessian, O. A. Wahab, A. Mourad, and H. Otrok, "Coalitional Federated Learning: Improving Communication and Training on Non-IID Data With Selfish Clients," *IEEE Transactions on Services Computing*, vol. 16, no. 4, pp. 2462–2476, July 2023.

[20] Z. Yi, Y. Jiao, W. Dai, G. Li, H. Wang, and Y. Xu, "A Stackelberg Incentive Mechanism for Wireless Federated Learning With Differential Privacy," *IEEE WIRELESS COMMUNICATIONS LETTERS*, vol. 11, no. 9, 2022.

[21] Y. Chen, H. Zhou, T. Li, J. Li, and H. Zhou, "Multifactor Incentive Mechanism for Federated Learning in IoT: A Stackelberg Game Approach," *IEEE Internet of Things Journal*, vol. 10, no. 24, pp. 21 595–21 606, Dec. 2023.

[22] M. B. Singh, H. Singh, and A. Pratap, "Stable Matching based Revenue Maximization for Federated Learning in UAV-Assisted WBANs," *IEEE Transactions on Services Computing*, pp. 1–12, 2024.

[23] Y. Zhan, P. Li, Z. Qu, D. Zeng, and S. Guo, "A learning-based incentive mechanism for federated learning," *IEEE Internet of Things Journal*, vol. 7, no. 7, pp. 6360–6368, 2020.

[24] X. Tang, H. Yu, X. Li, and S. Kraus, "Intelligent Agents for Auction-based Federated Learning: A Survey," Apr. 2024.

[25] U. Majeed, L. U. Khan, S. S. Hassan, Z. Han, and C. S. Hong, "FL-Incentivizer: FL-NFT and FL-Tokens for Federated Learning Model Trading and Training," *IEEE Access*, vol. 11, pp. 4381–4399, 2023.

[26] Y. Cui, L. Yao, Y. Li, Z. Chen, B. Ding, and X. Zhou, "An Auction-based Marketplace for Model Trading in Federated Learning," Feb. 2024.

[27] A. Yousafzai, L. U. Khan, U. Majeed, O. Hakeem, and C. S. Hong, "FedMarket: A Cryptocurrency Driven Marketplace for Mobile Federated Learning Services," *IEEE Access*, vol. 10, pp. 87 602–87 616, 2022.

[28] Z. Cheng, M. Liwang, X. Xia, M. Min, X. Wang, and X. Du, "Auction-Promoted Trading for Multiple Federated Learning Services in UAV-Aided Networks," June 2022.

[29] L. Dong and Y. Zhang, "Federated Learning Service Market: A Game Theoretic Analysis," in *2020 International Conference on Wireless Communications and Signal Processing (WCSP)*. IEEE, Oct. 2020, pp. 227–232.

[30] W. Cheng, Y. Zou, J. Xu, and W. Liu, "Dynamic Games for Social Model Training Service Market via Federated Learning Approach," *IEEE Transactions on Computational Social Systems*, vol. 9, no. 1, pp. 64–75, Feb. 2022.

[31] A. Ghosh, K. Ligett, A. Roth, and G. Schoenebeck, "Buying private data without verification," in *Proceedings of the fifteenth ACM conference on Economics and computation*, 2014, pp. 931–948.

[32] W. Wang, L. Ying, and J. Zhang, "Buying data from privacy-aware individuals: The effect of negative payments," in *Web and Internet Economics: 12th International Conference, WINE 2016, Montreal, Canada, December 11-14, 2016, Proceedings 12*. Springer, 2016, pp. 87–101.

[33] P. Dandekar, N. Fawaz, and S. Ioannidis, "Privacy auctions for recommender systems," *ACM Transactions on Economics and Computation (TEAC)*, vol. 2, no. 3, pp. 1–22, 2014.

[34] M. Zhang, F. Beltran, and J. Liu, "Selling Data at an Auction under Privacy Constraints," in *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence (UAI)*. PMLR, Aug. 2020, pp. 669–678.

[35] P. Duetting, Z. Feng, H. Narasimhan, D. Parkes, and S. S. Ravindranath, "Optimal Auctions through Deep Learning," in *Proceedings of the 36th International Conference on Machine Learning*. PMLR, May 2019, pp. 1706–1715.

[36] Z. Jorgensen, T. Yu, and G. Cormode, "Conservative or liberal? personalized differential privacy," in *2015 IEEE 31St international conference on data engineering*. IEEE, 2015, pp. 1023–1034.

[37] M. Bun and T. Steinke, "Concentrated differential privacy: Simplifications, extensions, and lower bounds," in *Theory of cryptography conference*. Springer, 2016, pp. 635–658.

[38] W. Cheng, Y. Zou, J. Xu, and W. Liu, "Dynamic games for social model training service market via federated learning approach," *IEEE Transactions on Computational Social Systems*, vol. 9, no. 1, pp. 64–75, 2021.

[39] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-iid data," *arXiv preprint arXiv:1806.00582*, 2018.

[40] K. Daskalakis, *The complexity of Nash equilibria*. University of California, Berkeley, 2008.

[41] J.-M. Lasry and P.-L. Lions, "Mean field games," *Japanese Journal of Mathematics*, vol. 2, no. 1, pp. 229–260, 2007.

[42] M. Huang, R. P. Malhamé, and P. E. Caines, "Large population stochastic dynamic games," *Communications in Information and Systems*, vol. 6, no. 3, pp. 221–252, 2006.

[43] F. Santambrogio, "Lecture notes on variational mean field games," in *Mean Field Games: Cetraro, Italy 2019*. Springer, 2021, pp. 159–201.

[44] P. Cardaliaguet, F. Delarue, J.-M. Lasry, and P.-L. Lions, *The Master Equation and the Convergence Problem in Mean Field Games*. Princeton University Press, 2019.

[45] P. Cardaliaguet, "Notes on mean field games," Lecture notes, Collège de France, 2013. [Online]. Available: https://www.ceremade.dauphine.fr/~cardaliaguet/MFG20130420.pdf

[46] N. Fournier and A. Guillin, "On the rate of convergence in wasserstein distance of the empirical measure," *Probability Theory and Related Fields*, vol. 162, no. 3, pp. 707–738, 2015.

X. Appendix

Theorem 1 (Privacy Auction Game Equilibrium) *In a PAG where privacy valuations v_i of DOs are drawn from joint distribution $\mathcal{W}$, the cost function is linear $c(v_i, \epsilon_i) = \epsilon_i \cdot v_i$, and the mechanism $\mathcal{M} = (q, p)$ satisfies DSIC, IR, DT, and BF, there exists a Bayesian-Nash Equilibrium in which all rational bidders truthfully report both their privacy valuations v_i* and *their privacy budgets ϵ_i:*

$$\forall i \in \mathcal{N}, \forall v_i \in \mathcal{V}_i, \quad \mathbb{E}[u_i(q, v_i)] \geq \mathbb{E}[u_i(q_{-i}, v_i)]. \tag{57}$$

Proof Sketch: Consider the threshold-based PAC mechanism (Algorithm 1), where valuations are sorted $v_1 \leq \cdots \leq v_n$, the largest k is selected such that $c(v_k, 1/(n-k)) \leq B/k$, and payments are $p_i = \min(B/k, c(v_{k+1}, 1/(n-k)))$ for $i \leq k$ and $p_i = 0$ otherwise.

DSIC (for $i \leq k$): Misreporting a higher v_i' may cause i to drop out of the top-k, yielding $p_i' = 0$ and $u_i' = 0$. Since $u_i = p_i - c(v_i, \epsilon_i) \geq 0$ (by IR), deviation yields no gain.

DSIC (for $i > k$): Misreporting a lower v_i' may enter the top-k, but the diluted payment $p_i' \leq B/(k+1)$ combined with the higher true cost $c(v_i, \epsilon_i') > c(v_i', \epsilon_i')$ results in $u_i' < 0$ or no improvement.

By monotonicity of the mechanism, misreporting cannot improve the allocation-payment tradeoff, so truthful reporting is a dominant strategy (DSIC) and all utilities are non-negative (IR).

BNE: By DSIC, for any fixed b_{-i}, truthful reporting maximises $u_i(v_i, \cdot)$. Taking expectations over $v_{-i} \sim \mathcal{W}_{-i}$ yields Eq. (57), which is the BNE definition.

For finite N, existence follows from Nash's theorem. In the large-N regime, the multi-agent game converges to a Mean-Field Game, where fixed-point theorems from MFG literature [44] guarantee equilibrium existence. ■

Theorem 2 (Existence of MFE) *Under Assumption 2, the privacy auction MFG admits at least one Mean-Field Equilibrium (Φ^*, μ^*).*

Proof: We apply Schauder's fixed-point theorem. Let $\mathcal{M} = \mathcal{C}([0, T]; \mathcal{P}_2(\mathcal{S}))$ with the sup-Wasserstein-1 metric. Define $\mathcal{T} : \mathcal{M} \to \mathcal{M}$ by $\mathcal{T}(\mu) = \hat{\mu}$, where $\hat{\mu}_t$ is the law of $\mathbf{s}(t)$ under the SDE of Eq. (22) driven by the optimal $b^*(\cdot, t; \mu)$ from Eq. (28). *Well-definedness* follows from viscosity solution theory under (R1)–(R3) [45]. *Continuity of* $\mathcal{T}$ follows from Lipschitz continuity of f and $\tilde{p}$ in μ (R1). *Relative compactness* follows from Prokhorov's theorem under (R2) and (R4). By Schauder's theorem, $\mathcal{T}$ has a fixed point $\mu^* = \mathcal{T}(\mu^*)$; setting $\Phi^* = \Phi(\cdot; \mu^*)$ yields the MFE. ■

Theorem 3 (MFG Approximation of Nash Equilibrium) *Let Assumptions 2 and 3 hold, and let (Φ^*, μ^*) be a MFE. Define the MFE-induced strategy profile:*

$$b_i^{N,*} = b^*(\mathbf{s}_i, t; \mu^*), \quad i = 1, \ldots, N. \tag{58}$$

Then $\mathbf{b}^{N,*} = (b_1^{N,*}, \ldots, b_N^{N,*})$ is an ε_N-Nash Equilibrium (Definition 5) of the PAG, where:

$$\varepsilon_N = \frac{2L_u C}{\sqrt{N-1}} = \mathcal{O}\left(\frac{1}{\sqrt{N}}\right), \tag{59}$$

and $C > 0$ depends only on the second moment of μ^* and the dimension of $\mathcal{S}$.

Proof: Fix arbitrary $i \in \mathcal{N}$ and $\hat{b}_i \in \mathcal{B}$.

Step 1 (Empirical measure decomposition). States $\{\mathbf{s}_j\}_{j \neq i}$ are i.i.d. from μ_0^* (MFE self-consistency). Define $\mu_{-i}^N = \frac{1}{N-1} \sum_{j \neq i} \delta_{\mathbf{s}_j}$. By the Wasserstein concentration inequality [46]:

$$\mathbb{E}\big[W_1(\mu_{-i}^N, \mu_0^*)\big] \leq \frac{C}{\sqrt{N-1}}. \tag{60}$$

Step 2 (Utility gap decomposition).

$$\begin{aligned} & u_i(\hat{b}_i, b_{-i}^{N,*}) - u_i(b_i^{N,*}, b_{-i}^{N,*}) \\ &= \underbrace{u_i(\hat{b}_i, b_{-i}^{N,*}) - u^{\mathrm{MFG}}(\mathbf{s}_i, \hat{b}_i; \mu_0^*)}_{\text{(I)}} \qquad (61) \\ &\quad + \underbrace{u^{\mathrm{MFG}}(\mathbf{s}_i, \hat{b}_i; \mu_0^*) - u^{\mathrm{MFG}}(\mathbf{s}_i, b_i^{N,*}; \mu_0^*)}_{\text{(II)}} \\ &\quad + \underbrace{u^{\mathrm{MFG}}(\mathbf{s}_i, b_i^{N,*}; \mu_0^*) - u_i(b_i^{N,*}, b_{-i}^{N,*})}_{\text{(III)}}. \qquad (62) \end{aligned}$$

Step 3 (Bound on (II)). Since $b_i^{N,*} = b^*(\mathbf{s}_i; \mu_0^*)$ maximises $u^{\mathrm{MFG}}(\mathbf{s}_i, \cdot\,; \mu_0^*)$: (II) ≤ 0.

Step 4 (Bound on (I) and (III)). Using Assumption 3: $|(\mathrm{I})| + |(\mathrm{III})| \leq 2L_u W_1(\mu_{-i}^N, \mu_0^*)$.

Step 5 (Conclusion). Taking expectations and applying the concentration bound:

$$\mathbb{E}\Big[u_i(\hat{b}_i, b_{-i}^{N,*}) - u_i(b_i^{N,*}, b_{-i}^{N,*})\Big] \leq \frac{2L_u C}{\sqrt{N-1}} = \varepsilon_N. \tag{63}$$

Since this holds for all i and all $\hat{b}_i$, $\mathbf{b}^{N,*}$ is an ε_N-Nash Equilibrium. ■